

FERMIONIC COHERENT STATES: CONVEX ORDER AND GAUSSIAN RANK

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ABSTRACT. We study two extremal properties of fermionic coherent states: concentration of their overlap distributions and length of exact Gaussian decompositions. For every basic exterior-power representation, the Husimi distribution of any density operator is dominated in convex order by that of a Slater state. For each strictly convex test function, equality characterizes pure Slater states. We prove the corresponding statements for the basic half-spin and odd-dimensional spin representations, obtain all positive-order Rényi–Wehrl minima, and evaluate the coherent distributions and entropy constants. The proof uses exact conditioning and a single convex decomposition shared by all measured directions. For Gaussian rank, a degree-eighteen relative invariant excludes every three-term decomposition of two copies of the four-mode magic state. This yields rank multiplicativity for two arbitrary nonzero four-mode even vectors. The exterior-power convex-order theorem, its equality cases, and the entropy minimizer theorems have end-to-end Lean proofs. The formalization of the Gaussian-rank theorem currently includes the original rank bounds two and four and the full normal-family polynomial certificate.

1. INTRODUCTION

A fermionic state can be compared with simple states in two different ways. One can measure its overlaps with them, or ask how many are needed to express it as a sum. The first question concerns a probability distribution; the second concerns an exact vector identity. This paper treats both questions using the geometry of Slater and Gaussian states.

For a fixed number of particles, the one-particle space is $\mathbb{C}^n$ and the p -particle space is its exterior power $\Lambda^p \mathbb{C}^n$. A unit vector $u_1 \wedge \cdots \wedge u_p$, with the u_i orthonormal, is called a *Slater vector*. It describes the occupation of p mutually orthogonal one-particle modes. Even in a small space, most vectors need not have this form. For example, in $\Lambda^2 \mathbb{C}^4$ the normalized vector

$$(1.1) \quad \psi = \frac{e_1 \wedge e_2 + e_3 \wedge e_4}{\sqrt{2}}$$

is not a Slater vector: its exterior square is nonzero, whereas the square of a decomposable two-vector vanishes. Its overlap with $e_1 \wedge e_2$ has squared modulus $1/2$. Rotating the reference Slater vector gives an entire distribution of such overlaps.

A density operator is a positive semidefinite operator ρ of trace one; it includes both pure states $|\psi\rangle\langle\psi|$ and mixtures. If Ω_x is a unit coherent vector indexed by a point x of the coherent orbit, its *Husimi function* is

$$Q_\rho(x) = \langle \Omega_x, \rho \Omega_x \rangle.$$

The inner product is conjugate-linear in its first argument. The orbit carries the probability measure ν induced by normalized Haar measure of the acting compact group. Thus Q_ρ is a random variable in $[0, 1]$. For a D -dimensional irreducible representation, its mean is $1/D$. For a unit input vector ψ , we abbreviate $Q_\psi = Q_{|\psi\rangle\langle\psi|}$.

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1.1. Comparing the whole distribution. Our distribution theorem says that coherent inputs maximize every continuous convex average:

$$(1.2) \quad \int \Phi(Q_\rho) d\nu \leq \int \Phi(Q_{P_\Omega}) d\nu, \quad P_\Omega = |\Omega\rangle\langle\Omega|.$$

Here Ω is any coherent unit vector, and Φ is continuous and convex on $[0, 1]$. This relation is called *convex order*. Since all these random variables have the same mean, it says that the coherent distribution has the greatest spread in every convex test. It implies a comparison of variances by taking $\Phi(t) = t^2$, and of Wehrl entropies by taking $\Phi(t) = t \log t$ with the continuous value zero at $t = 0$. For every strictly convex Φ , equality in (1.2) forces the whole input density to be a pure coherent state.

We prove this statement for all basic exterior powers (Theorem 2.1), for both basic half-spin representations (Theorem 3.1), and for the basic odd-dimensional spin representation (Corollary 3.5). In the half-spin realization used below, the coherent vectors are pure fermionic Gaussian vectors, whose particle number may vary while their parity remains fixed. Section 3 constructs this realization and explains its relation to odd-dimensional spin.

The entropy problem has a long history. Gnutzmann and Życzkowski introduced Rényi–Wehrl localization measures in the generalized coherent state setting [1]. Sugita proved coherent-state optimality for integer indices greater than one [2], and studied the resulting moments for many-particle states [3]. Lieb and Solovej proved a convex-function comparison for symmetric $SU(N)$ coherent states [4]. Symmetric powers describe a different representation series from the exterior powers considered here. Our comparison treats arbitrary continuous convex functions and all mixed inputs in the stated fermionic series. In particular, it yields all real positive-order Rényi–Wehrl minima, together with their equality cases.

The main step is a compatibility statement about conditioning. Detecting one occupied mode turns the original density into a density on a smaller fermionic space. Its normalization is an occupation probability, determined by a one-particle matrix or a Majorana covariance matrix. We express this matrix as a convex combination of coherent matrices using one set of probabilities, chosen from the diagonal of the same rotated full density. These probabilities work for every subsequent measured direction. The conditional coherent states also remain coherent in the smaller space. The induction therefore uses the same comparison state at every stage. The equality argument recovers the full density from this common decomposition; it does not identify a many-particle state merely by its one-particle matrix.

1.2. Counting Gaussian summands. Allowing superpositions of particle numbers leads to a different question. The *Gaussian rank* $\chi_G(\psi)$ is the least number of nonzero Gaussian vectors whose sum is ψ , with arbitrary complex coefficients absorbed into the summands. The definition and parity convention appear in Section 5.

On four modes, let $|0000\rangle$ and $|1111\rangle$ denote the vacuum and the state with all four modes occupied, and put

$$M = \frac{|0000\rangle + |1111\rangle}{\sqrt{2}}.$$

This state is a basic non-Gaussian resource in matchgate computation. Its two-copy expansion has four Gaussian terms. A shorter expansion, however, could use terms coupling the two blocks of four modes. Cudby and Strelchuk proved a rank-four result under symmetry restrictions and formulated the unrestricted two-copy problem [5]. Gaussian decompositions are also central to the simulation methods of Dias and Koenig [6]. Related multiplicativity results for Gaussian fidelity and extent concern different optimization problems [7]. Recent work of Tarabunga studies non-Gaussianity through the eigenvalue distribution of a two-copy Majorana operator and its behavior under postselected Gaussian operations [8].

We prove that $\chi_G(M^{\otimes 2}) = 4$, and more generally that Gaussian rank is multiplicative for two arbitrary nonzero four-mode even vectors. The obstruction is a polynomial built from the full bilinear Gram of the Clifford bivector action. It is nonzero at the target and vanishes on every sum of three pure spinors. A common change of coordinates first normalizes two summands. A polynomial deformation then brings the third summand into a tractable chart, and a specialization argument recovers all degenerate cases. The change of coordinates acts on the entire sum: transforming the summands independently would destroy the equation being tested.

1.3. Formal verification and organization. The accompanying Lean development verifies the exterior-power convex-order theorem, the actual Slater-orbit probability measure, the strict equality characterization, and the positive-order moment and entropy minimizer theorems. These are end-to-end proofs with no additional mathematical hypotheses beyond those stated in the theorems. The standard logical foundations and the precise boundary of the formalization are recorded in Section 7. In particular, the Gaussian-rank four conclusion still has a paper proof supplemented by verified coordinate certificates and original-object lemmas.

The sources, full declaration types, and verification evidence are available at

<https://github.com/veridiscoversLab/fermionic-coherent-lean>.

We begin with exterior powers, where the conditioning argument is most transparent, and then pass to spin representations. After evaluating the coherent distributions and entropy constants, we turn to Gaussian rank. The final application section gives quantitative entropy bounds and the information capacity of the specified coherent measurements.

2. EXTERIOR-POWER HUSIMI CONVEX ORDER

2.1. Slater states and the measurement. Let $E = \mathbb{C}^n$ with its standard Hermitian inner product, linear in the second variable. For $0 \leq p \leq n$, equip

$$\mathcal{H}_{p,n} = \Lambda^p E, \quad D_{p,n} = \dim \mathcal{H}_{p,n} = \binom{n}{p},$$

with the inner product for which increasing wedges of an orthonormal basis are orthonormal. Thus a wedge of p orthonormal vectors has norm one. The induced unitary representation is $R_p(U) = \Lambda^p U$, for $U \in U(n)$.

If $W \subset E$ is a p -dimensional subspace and $(w_1, \dots, w_p)$ is an orthonormal basis of W , the vector $\Omega_W = w_1 \wedge \dots \wedge w_p$ is a *unit Slater vector*. A different orthonormal basis changes it by a scalar of modulus one. Consequently,

$$\rho_W = |\Omega_W\rangle\langle\Omega_W|$$

is determined by W . We call ρ_W a *pure Slater density* and write π_W for the orthogonal projection from E onto W ; these operators act on different spaces. The Slater-projector orbit is $\mathcal{O}_{p,n} = \{\rho_W : \dim W = p\}$.

Fix a unit Slater reference Ω_0 . The pushforward of Haar probability measure on $U(n)$ under

$$U \mapsto |R_p(U)\Omega_0\rangle\langle R_p(U)\Omega_0|$$

is denoted by $\nu_{p,n}$. Transitivity and Haar invariance make this measure independent of the reference. Equivalently, it is the invariant probability measure on the Grassmannian of p -planes, expressed through their Slater projectors.

A *density operator* on $\mathcal{H}_{p,n}$ is an operator $\rho \geq 0$ with $\text{Tr } \rho = 1$. Its Husimi function is

$$(2.1) \quad Q_\rho(P) = \text{Tr}(\rho P), \quad P \in \mathcal{O}_{p,n}.$$

We also write $Q_\rho(U) = \langle R_p(U)\Omega_0, \rho R_p(U)\Omega_0 \rangle$ for the same measurement pulled back to the group. Since $0 \leq \rho \leq \text{Id}$, its values lie in $[0, 1]$.

The normalization is

$$(2.2) \quad \int_{\mathcal{O}_{p,n}} P d\nu_{p,n}(P) = \frac{\text{Id}_{\mathcal{H}_{p,n}}}{D_{p,n}}, \quad \int Q_\rho d\nu_{p,n} = \frac{1}{D_{p,n}}.$$

Indeed, the averaged projector commutes with every $R_p(U)$. Commutation with diagonal one-particle unitaries makes it diagonal in the occupation basis, because distinct occupied subsets have distinct characters. Permutation unitaries make all its diagonal entries equal. Its trace is one, which proves the first identity; pairing with ρ proves the second. For $p = 0$ and $p = n$, the sector is one-dimensional, its unique density is Slater, and $Q_\rho \equiv 1$. This also includes $n = p = 0$.

Theorem 2.1 (Exterior-power Husimi convex order). *For every $0 \leq p \leq n$, every density operator ρ on $\mathcal{H}_{p,n}$, every continuous convex function $\Phi : [0, 1] \rightarrow \mathbb{R}$, and every pure Slater density ρ_W ,*

$$(2.3) \quad \int \Phi(Q_\rho) d\nu_{p,n} \leq \int \Phi(Q_{\rho_W}) d\nu_{p,n}.$$

The right-hand side is independent of W . If Φ is strictly convex, equality holds if and only if ρ is a pure Slater density.

All the compared distributions have the mean in (2.2). The theorem says that a Slater input maximizes every continuous convex observation of the measurement value. The equality statement concerns the whole density operator, including its off-diagonal entries.

2.2. A shared convex mixture. We first record the probabilistic step used here and in the spinor argument. Its coefficients are fixed before the measurement variable is sampled.

Lemma 2.2 (Shared-mixture comparison). *Let (X, μ) be a probability space. Suppose measurable functions $a, a_1, \dots, a_L : X \rightarrow [0, 1]$ satisfy*

$$a(x) = \sum_{j=1}^L t_j a_j(x), \quad t_j \geq 0, \quad \sum_{j=1}^L t_j = 1,$$

and all a_j have the same distribution η . For every continuous convex function $H : [0, 1] \rightarrow \mathbb{R}$,

$$(2.4) \quad \int_X H(a(x)) d\mu(x) \leq \int_{[0,1]} H(t) d\eta(t).$$

If H is strictly convex, equality implies that all the a_j with $t_j > 0$ agree almost everywhere.

Proof. Pointwise Jensen gives $H(a) \leq \sum_j t_j H(a_j)$. Integration and the common distribution give (2.4). All functions being integrated are bounded. If the integrals are equal, the nonnegative Jensen difference vanishes almost everywhere. The equality criterion for strict convexity gives the assertion. $\square$

If R is a $[0, 1]$ -valued random variable, define

$$(2.5) \quad H_\Phi(t) = \mathbb{E}\Phi(tR), \quad 0 \leq t \leq 1.$$

Continuity follows by bounded convergence, and convexity follows pointwise. If Φ is strictly convex and $\Pr(R > 0) > 0$, then H_Φ is strictly convex: for distinct s, t the strict scalar inequality holds on the event $R > 0$.

2.3. Contraction and the one-particle density. For $u \in E$, let $c(u)^*$ denote creation by u , namely $\xi \mapsto u \wedge \xi$, and let $c(u)$ be its adjoint. Thus $c(u)$ is conjugate-linear in u . These operators act on the full graded exterior space $\Lambda^\bullet E$ and change degree by one; contraction vanishes on degree zero. On a decomposable vector,

$$(2.6) \quad c(u)(v_1 \wedge \cdots \wedge v_p) = \sum_{j=1}^p (-1)^{j-1} \langle u, v_j \rangle v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_p.$$

The hat indicates the omitted factor. The canonical anticommutation relations are

$$c(u)c(v) + c(v)c(u) = 0, \quad c(u)c(v)^* + c(v)^*c(u) = \langle u, v \rangle \text{Id}.$$

For unit u , the operator $N_u = c(u)^*c(u)$ is the projection onto the states in which u is occupied. If $(e_i)_{i=1}^n$ is an orthonormal basis, then on $\mathcal{H}_{p,n}$,

$$(2.7) \quad \sum_{i=1}^n N_{e_i} = p \text{Id}.$$

Both assertions can be read directly in the occupation basis.

For a density ρ on $\mathcal{H}_{p,n}$, define its one-particle density γ_ρ by

$$(2.8) \quad a_\rho(u) := \text{Tr}(c(u)\rho c(u)^*) = \text{Tr}(\rho N_u) = \langle u, \gamma_\rho u \rangle.$$

Polarization gives a unique Hermitian operator on E . The projection property of N_u and (2.7) imply

$$(2.9) \quad 0 \leq \gamma_\rho \leq \text{Id}_E, \quad \text{Tr } \gamma_\rho = p.$$

For a Slater input, $\gamma_{\rho_W} = \pi_W$.

Lemma 2.3 (One-particle mixture from the original density). *Choose an orthonormal eigenbasis $(f_i)_{i=1}^n$ of γ_ρ . For every p -element subset $S \subset \{1, \dots, n\}$, let Ω_S be its increasing occupation wedge, $W_S = \text{span}\{f_i : i \in S\}$, and $t_S = \langle \Omega_S, \rho \Omega_S \rangle$. Then*

$$(2.10) \quad t_S \geq 0, \quad \sum_{|S|=p} t_S = 1, \quad \gamma_\rho = \sum_{|S|=p} t_S \pi_{W_S}.$$

If $n \geq 1$, then for a uniformly distributed unit vector $u \in E$, any continuous convex $H : [0, 1] \rightarrow \mathbb{R}$, and any rank- p projection π_W ,

$$(2.11) \quad \mathbb{E}_u H(\langle u, \gamma_\rho u \rangle) \leq \mathbb{E}_u H(\langle u, \pi_W u \rangle).$$

If H is strictly convex, equality forces exactly one t_S to be positive, and consequently $\rho = |\Omega_S\rangle\langle\Omega_S|$.

Proof. The t_S are diagonal entries of the same positive trace-one operator in a complete orthonormal basis, so they are probabilities. For each i , the occupation operator N_{f_i} is diagonal in this basis and has eigenvalue 1 precisely when $i \in S$. Hence

$$\langle f_i, \gamma_\rho f_i \rangle = \text{Tr}(\rho N_{f_i}) = \sum_{S \ni i} t_S.$$

The off-diagonal entries of γ_ρ vanish in the chosen eigenbasis. This proves (2.10). In particular, for every unit u at once,

$$a_\rho(u) = \sum_{|S|=p} t_S \langle u, \pi_{W_S} u \rangle.$$

The projections on the right have the same rank, so their quadratic readouts have the same spherical distribution. Lemma 2.2 gives (2.11).

For $S \neq T$, the Hermitian operator $\pi_{W_S} - \pi_{W_T}$ is nonzero. Its quadratic form is nonzero somewhere on the unit sphere, and therefore on a nonempty open set of positive spherical measure.

The strict equality criterion excludes two such positive-weight indices. The remaining diagonal entry of ρ is one. Positivity gives

$$|\langle \Omega_S, \rho \Omega_T \rangle|^2 \leq t_S t_T$$

for all occupied subsets, so every other matrix entry is zero. Thus the original density is the stated pure Slater projector. $\square$

Remark 2.4 (The Pauli convex set). The set of Hermitian matrices satisfying (2.9) is the convex hull of rank- p orthogonal projections. To see this, diagonalize such a matrix. The vertices of $\{x \in [0, 1]^n : \sum_i x_i = p\}$ are the zero-one vectors with p ones: a nonintegral point has at least two nonintegral coordinates, which admit opposite small perturbations preserving the sum. Rotating a vertex decomposition back proves the assertion. For the density in the theorem, (2.10) supplies this decomposition directly from its occupation probabilities and also yields the full-density equality criterion.

2.4. The conditional measurement. For a unit vector u , put $A_\rho(u) = c(u)\rho c(u)^*$. It is positive with trace $a_\rho(u)$. The orthogonal splitting

$$\Lambda^{p-1}E = \Lambda^{p-1}(u^\perp) \oplus u \wedge \Lambda^{p-2}(u^\perp)$$

shows that $\ker c(u)$ in degree $p-1$ is $\Lambda^{p-1}(u^\perp)$; the second summand is absent when $p=1$. Since $c(u)^2 = 0$, the range of $A_\rho(u)$ lies in this kernel. If $a_\rho(u) > 0$, then

$$(2.12) \quad \rho_u = \frac{A_\rho(u)}{a_\rho(u)}$$

is a density on $\Lambda^{p-1}(u^\perp)$. For a unit Slater vector ξ in that space, adjointness gives

$$(2.13) \quad \langle u \wedge \xi, \rho(u \wedge \xi) \rangle = \langle \xi, A_\rho(u)\xi \rangle = a_\rho(u) \langle \xi, \rho_u \xi \rangle.$$

If $a_\rho(u) = 0$, positivity and zero trace give $A_\rho(u) = 0$, so the first expression in (2.13) is zero as well.

The relevant Haar decomposition can be written without choosing a global frame for the bundle $u^\perp$. For $p \geq 1$, take $\Omega_0 = e_1 \wedge \cdots \wedge e_p$ and embed $U(n-1)$ into $U(n)$ by $\iota(V) = 1 \oplus V$. Right invariance gives

$$(2.14) \quad \int_{U(n)} \Phi(Q_\rho(U)) dU = \int_{U(n)} \int_{U(n-1)} \Phi(Q_\rho(U\iota(V))) dV dU.$$

For fixed U , the first vector is $u = Ue_1$, and the remaining wedge is a Haar Slater vector in $\Lambda^{p-1}(u^\perp)$. The outer variable u is uniform on the complex unit sphere. Thus (2.13) applies to the same measurement appearing in (2.14). In particular, the outer measure remains Haar probability measure; $a_\rho(u)$ enters the argument of Φ .

Lemma 2.5 (Slater closure under contraction). *If $a_{\rho_W}(u) > 0$, the conditional density $(\rho_W)_u$ is a pure Slater density with occupied space $W \cap u^\perp$.*

Proof. Here $a_{\rho_W}(u) = \|\pi_W u\|^2$. Choose an orthonormal basis of W whose first vector is $w_1 = \pi_W u / \|\pi_W u\|$. Its other vectors $w_2, \dots, w_p$ lie in $u^\perp$. Up to the phase of Ω_W , formula (2.6) gives

$$c(u)\Omega_W = \|\pi_W u\| w_2 \wedge \cdots \wedge w_p.$$

Normalizing its rank-one projector proves the assertion, including the vacuum conditional state when $p=1$. $\square$

2.5. Induction and equality.

Proof of Theorem 2.1. We induct on p , simultaneously for all $n \geq p$. The case $p = 0$ has $Q \equiv 1$. Suppose $p \geq 1$ and the assertion is known in degree $p - 1$. Let R have the Husimi distribution of a Slater input in $\mathcal{H}_{p-1, n-1}$ and set $H_\Phi(t) = \mathbb{E}\Phi(tR)$. This function is continuous and convex. If Φ is strictly convex, then H_Φ is strictly convex, since

$$\mathbb{E}R = \binom{n-1}{p-1}^{-1} > 0$$

by (2.2).

For every positive-weight conditional density (2.12), apply the induction hypothesis with the test function $x \mapsto \Phi(a_\rho(u)x)$. Zero-weight branches give $\Phi(0)$ on both sides. Integrating (2.14), and then applying Lemma 2.3, yields

$$\begin{aligned} \int \Phi(Q_\rho) d\nu_{p,n} &\leq \mathbb{E}_u H_\Phi(\langle u, \gamma_\rho u \rangle) \\ (2.15) \qquad &\leq \mathbb{E}_u H_\Phi(\langle u, \pi_W u \rangle) = \int \Phi(Q_{\rho_W}) d\nu_{p,n}. \end{aligned}$$

The final equality follows from Lemma 2.5: each nonzero conditional Slater input has the same lower-dimensional Husimi distribution. The original double-Haar integrands are continuous on compact groups. The normalized conditional matrices are measurable where their continuous weights are positive, and the zero branch was handled separately. Boundedness therefore justifies these integrations and their order changes.

If Φ is strictly convex and the first and last quantities in (2.15) are equal, the middle Jensen comparison is an equality. Lemma 2.3, applied to the strictly convex H_Φ , forces ρ to be a pure Slater density. Conversely, all Slater inputs have the same Husimi law by Haar invariance and hence attain equality. The filled sector $p = n$ is one-dimensional and agrees with this induction. $\square$

Example 2.6 (One particle). Every unit vector in E is a one-particle Slater vector. In two modes, a density with eigenvalues t and $1 - t$ has measurement

$$Q_\rho(u) = t|u_1|^2 + (1 - t)|u_2|^2.$$

At $t = 1/2$ this is the constant $1/2$; at $t = 1$ it has the distribution of $|u_1|^2$. Both have mean $1/2$. Theorem 2.1 compares all continuous convex observations between these cases and every intermediate density. Its strict equality criterion singles out the rank-one inputs.

Example 2.7 (Two-particle coherence). In $\Lambda^2 \mathbb{C}^4$, write $e_{ij} = e_i \wedge e_j$ and consider

$$\psi = \frac{e_{12} + e_{34}}{\sqrt{2}}, \quad \rho_{\text{mix}} = \frac{1}{2}|e_{12}\rangle\langle e_{12}| + \frac{1}{2}|e_{34}\rangle\langle e_{34}|.$$

Both $|\psi\rangle\langle\psi|$ and ρ_{mix} have one-particle density $\frac{1}{2}\text{Id}_E$. In the standard eigenbasis, the shared occupation mixture places weight $1/2$ on each of $\{1, 2\}$ and $\{3, 4\}$. Their full Husimi functions differ: at the plane spanned by $(e_1 + e_3)/\sqrt{2}$ and $(e_2 + e_4)/\sqrt{2}$, their values are respectively $1/2$ and $1/4$. Thus the one-particle mixture retains exactly the information needed for the common conditional-weight comparison, while the conditional density in (2.12) still carries the original coherence. Both inputs give strict inequality in (2.3) for every strictly convex test.

3. HUSIMI CONVEX ORDER FOR SPIN REPRESENTATIONS

An occupation measurement removes one fermionic mode. Its probability is an affine function of the Majorana covariance matrix, while its conditional state lives in a smaller spin representation. Two facts make this reduction compatible with convex order: one probability mixture represents the occupation probabilities in every rotated mode, and a nonzero occupation projection of a pure

Gaussian state remains Gaussian. We prove both facts for the original Fock-space action before applying the conditional comparison.

3.1. Fock space, Majorana operators, and coherent rays. Let

$$\mathcal{F}_n = \Lambda^\bullet \mathbb{C}^n, \quad \mathcal{F}_n^\epsilon = \bigoplus_{k \equiv \epsilon \pmod{2}} \Lambda^k \mathbb{C}^n, \quad \epsilon \in \{0, 1\}.$$

The increasing occupation wedges form an orthonormal basis, denoted by $|s\rangle$, $s \in \{0, 1\}^n$. Creation is exterior multiplication, $a_j^* \xi = e_j \wedge \xi$, and annihilation a_j is its Hilbert adjoint. Thus $a_j a_k + a_k a_j = 0$ and $a_j a_k^* + a_k^* a_j = \delta_{jk} \text{Id}$. Define

$$(3.1) \quad c_{2j-1} = a_j + a_j^*, \quad c_{2j} = i(a_j^* - a_j).$$

These Majorana operators are self-adjoint and satisfy

$$(3.2) \quad c_\alpha c_\beta + c_\beta c_\alpha = 2\delta_{\alpha\beta} \text{Id}.$$

For $z \in \mathbb{C}^{2n}$ write $c(z) = \sum_\alpha z_\alpha c_\alpha$. The quadratic form relevant to (3.2) is the complex *bilinear* form $b(z, w) = \sum_\alpha z_\alpha w_\alpha$.

We use the positive real Clifford realization of $\text{Spin}(2n)$: it consists of products of an even number of real unit Clifford vectors. Its Fock representation sends such a product to the corresponding product of the $c(v)$, and is denoted by $g \mapsto U_g$. Each factor $c(v)$ is unitary and reverses parity; hence U_g is unitary and preserves $\mathcal{F}_n^\epsilon$. Conjugation gives

$$(3.3) \quad U_g c(v) U_g^* = c(R_g v), \quad R_g \in \text{SO}(2n).$$

The map onto $\text{SO}(2n)$ is surjective. Indeed, $c(w)c(v)c(w) = c(2\langle w, v \rangle w - v)$ for a real unit w ; products of two such reflections give the plane rotations generating $\text{SO}(2n)$. Allowing odd products realizes the other component of $O(2n)$. An implementing word has odd length precisely when the orthogonal transformation reverses orientation. These elementary Clifford facts also describe the parity change of its Fock action.

A *pure Gaussian ray* in $\mathcal{F}_n^\epsilon$ means a ray in the $\text{Spin}(2n)$ orbit of an occupation vector of parity ϵ . All occupation vectors of that parity belong to the same orbit: changing an even number of occupation bits is effected, up to a phase, by the corresponding product of Majorana operators. Choose a unit representative $\Omega_{n,\epsilon}$ of this orbit. These are the coherent rays of the basic half-spin representation; see also [9] for their geometric description.

Each parity representation is irreducible. To see this directly, the occupation projections

$$|s\rangle\langle s| = \prod_{j:s_j=1} a_j^* a_j \prod_{j:s_j=0} (\text{Id} - a_j^* a_j)$$

belong to the complex span of even Majorana words. Multiplying them by even words which change the occupied bits gives every matrix unit $|t\rangle\langle s|$ with $|s| \equiv |t| \pmod{2}$. Thus the algebra spanned by the representation acts as the full matrix algebra on each sector. In particular, with $D = 2^{n-1}$ and normalized Haar measure dg ,

$$(3.4) \quad \int_{\text{Spin}(2n)} |U_g \Omega_{n,\epsilon}\rangle \langle U_g \Omega_{n,\epsilon}| dg = \frac{\text{Id}_{\mathcal{F}_n^\epsilon}}{D}.$$

The average commutes with the representation and has trace one, which proves the formula. For a density operator ρ on this sector let

$$Q_\rho(g) = \langle U_g \Omega_{n,\epsilon}, \rho U_g \Omega_{n,\epsilon} \rangle.$$

Consequently $0 \leq Q_\rho \leq 1$ and $\int Q_\rho dg = D^{-1}$. All parity indices below are interpreted modulo two.

Theorem 3.1 (Half-spin Husimi convex order). *Let $n \geq 1$ and $\epsilon \in \{0, 1\}$. For every density operator ρ on $\mathcal{F}_n^\epsilon$, every continuous convex function $\Phi : [0, 1] \rightarrow \mathbb{R}$, and every unit pure Gaussian vector $\Omega \in \mathcal{F}_n^\epsilon$,*

$$(3.5) \quad \int_{\text{Spin}(2n)} \Phi(Q_\rho(g)) dg \leq \int_{\text{Spin}(2n)} \Phi(Q_{|\Omega\rangle\langle\Omega|}(g)) dg.$$

The right-hand side is independent of Ω . If Φ is strictly convex, equality holds if and only if ρ is a pure Gaussian projector.

3.2. One covariance decomposition for all rotated modes. The first-mode occupation projection and the Majorana covariance are

$$(3.6) \quad P = a_1^* a_1 = \frac{\text{Id} + i c_1 c_2}{2}, \quad (\Gamma_\rho)_{\alpha\beta} = \frac{i}{2} \text{Tr}(\rho[c_\alpha, c_\beta]).$$

The matrix Γ_ρ is real and skew-symmetric, and $\Gamma_{U_g \rho U_g^*} = R_g \Gamma_\rho R_g^\top$. Putting $u = R_g e_1$ and $v = R_g e_2$, we have

$$(3.7) \quad a_\rho(g) := \text{Tr}(P U_g^* \rho U_g) = \frac{1 + u^\top \Gamma_\rho v}{2}.$$

Thus every first-mode probability is a readout of the same covariance.

Lemma 3.2 (Covariance mixture within one parity orbit). *For any density operator ρ on $\mathcal{F}_n^\epsilon$ there are unit pure Gaussian vectors χ_s of parity ϵ and numbers $t_s \geq 0$, $\sum_s t_s = 1$, such that*

$$(3.8) \quad \Gamma_\rho = \sum_s t_s \Gamma_{\chi_s}, \quad a_\rho(g) = \sum_s t_s a_{\chi_s}(g) \quad \text{for every } g \in \text{Spin}(2n).$$

The probabilities t_s are independent of g .

Proof. Choose $R \in \text{SO}(2n)$ such that

$$R \Gamma_\rho R^\top = \text{diag}(\lambda_1 J, \dots, \lambda_n J), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The usual real skew-symmetric normal form gives an orthogonal R . If its determinant is negative, reversing one coordinate changes the sign of one λ_j and makes R special orthogonal. The signed parameters allow this adjustment within the given parity sector. Choose a Spin lift and let σ be the correspondingly rotated density operator. Set $t_s = \langle s, \sigma s \rangle$. Positivity and trace one give a probability distribution supported on $|s| \equiv \epsilon \pmod{2}$.

For an occupation vector,

$$\Gamma_{|s\rangle} = \text{diag}((2s_1 - 1)J, \dots, (2s_n - 1)J).$$

The diagonal two-by-two blocks of Γ_σ are determined by the occupation expectations $\text{Tr}(\sigma a_j^* a_j)$; its other blocks vanish by construction. It follows that $\Gamma_\sigma = \sum_s t_s \Gamma_{|s\rangle}$. Rotating the occupation vectors back gives the first identity in (3.8). Equation (3.7) then gives the second identity simultaneously for every g . $\square$

This lemma decomposes a covariance matrix. The many-body operator ρ itself need not be a mixture of Gaussian projectors. What enters the convex comparison is the common set of probabilities in (3.8).

3.3. Pure spinors and conditional Gaussian closure. For a nonzero $\chi \in \mathcal{F}_n$ define its Clifford annihilator by

$$(3.9) \quad L_\chi = \{z \in \mathbb{C}^{2n} : c(z)\chi = 0\}.$$

Equation (3.2) implies that L_χ is isotropic for b , so $\dim L_\chi \leq n$. A spinor with $\dim L_\chi = n$ is called *pure*. This is an algebraic condition on the original Fock vector. For conditioning at the last mode, we use $\mathcal{F}_0^0 = \mathbb{C}$, $\mathcal{F}_0^1 = 0$ and call the unique nonzero zero-mode ray Gaussian.

Lemma 3.3 (Gaussian orbit and occupation conditioning). *The pure-spinor rays in $\mathcal{F}_n^\epsilon$ are exactly its pure Gaussian rays. If $b(z, z) = 0$, χ is pure, and $c(z)\chi \neq 0$, then $c(z)\chi$ is pure. In particular, if $P\chi \neq 0$, then under the occupation-space identification $P\mathcal{F}_n^\epsilon = |1\rangle \otimes \mathcal{F}_{n-1}^{\epsilon-1}$,*

$$(3.10) \quad \frac{P\chi}{\|P\chi\|} = |1\rangle \otimes \phi$$

for a unit pure Gaussian vector ϕ in the remaining $n - 1$ modes. For the vacant projection $\text{Id} - P$, the same conclusion holds with first factor $|0\rangle$ and remaining parity ϵ .

Proof. First let $L \subset \mathbb{C}^{2n}$ be maximal isotropic. It has trivial intersection with $\bar{L}$: a nonzero z in that intersection would satisfy $z, \bar{z} \in L$ and $b(z, \bar{z}) = \sum_\alpha |z_\alpha|^2 > 0$. Choose a Hermitian orthonormal basis $z_j = (p_j + iq_j)/\sqrt{2}$ of L , with p_j, q_j real. The equations $b(z_j, z_k) = 0$ and $\sum_\alpha \overline{(z_j)_\alpha} (z_k)_\alpha = \delta_{jk}$ show that $p_1, q_1, \dots, p_n, q_n$ form a real orthonormal frame. An orthogonal transformation carries the standard frame to this one. Implement it by a product W of real unit Majorana operators, as above. It carries $\text{Span}\{e_{2j-1} + ie_{2j} : 1 \leq j \leq n\}$ to L . The corresponding standard Clifford operators are $2a_j$.

Their joint kernel is precisely the vacuum line: in an occupation expansion, $a_j\xi = 0$ removes every coefficient of a wedge containing e_j , and requiring this for all j leaves only the scalar wedge. Conjugating by W proves that the joint kernel associated with L is the single line $\mathbb{C}W|0\rangle$. Its parity is the length of the implementing word modulo two. Thus two such lines of the same parity are related by an even word, hence by $\text{Spin}(2n)$. Conversely, the vacuum has an n -dimensional annihilator, and orthogonal conjugation preserves that dimension. This proves the equality of the pure-spinor and Gaussian ray sets in each sector, including the odd reference orbit.

Now suppose $c(z)\chi \neq 0$ and $b(z, z) = 0$. Then $z \notin L_\chi = L_\chi^\perp$, so $K = L_\chi \cap z^\perp$ has dimension $n - 1$. For $w \in K$, the Clifford relation gives $c(w)c(z)\chi = 0$, and $c(z)^2 = 0$ gives the same conclusion for $w = z$. Therefore the annihilator of $c(z)\chi$ contains $K \oplus \mathbb{C}z$, an n -dimensional isotropic space. The resulting spinor is pure.

Both a_1 and a_1^* are nonzero scalar multiples of isotropic Clifford vectors. If $P\chi = a_1^*a_1\chi \neq 0$, both successive actions are nonzero, so $P\chi$ is pure. Write its normalization as in (3.10). Its annihilator contains the a_1^* direction. Isotropy forces every annihilating vector to have zero a_1 coefficient. After subtracting its a_1^* component, the remaining vector lies in the last $2n - 2$ Majorana directions. Modulo the a_1^* line, the annihilator therefore gives an $(n - 1)$ -dimensional isotropic annihilator of ϕ . The first part of the proof identifies ϕ as Gaussian. Removing one occupied mode changes its parity from ϵ to $\epsilon - 1$. For a nonzero vacant projection, use $\text{Id} - P = a_1a_1^*$ and reverse the two null-vector actions. Its annihilator contains the a_1 direction; quotienting by that line proves the same statement for the remaining modes, now without a parity change. $\square$

3.4. Haar conditioning and strict equality.

Proof of Theorem 3.1. We induct on n , simultaneously for the two parity sectors. For $n = 1$ each sector is one-dimensional and the assertion is immediate. For $n \geq 2$ we may choose the reference ray in the form

$$\Omega_{n,\epsilon} = |1\rangle \otimes \Omega_{n-1,\epsilon-1}.$$

Let $G_n = \text{Spin}(2n)$ and let $H_n = \text{Spin}(2n - 2)$ be the subgroup generated by the last $2n - 2$ Majorana directions. Its action on the occupied first-mode subspace is $\text{Id} \otimes U_h$. Indeed, every remaining odd Majorana acquires a first-mode parity sign in the tensor identification, and the signs cancel in each even word defining H_n . Right invariance of Haar measure gives

$$(3.11) \quad \int_{G_n} \Phi(Q_\rho(g)) dg = \int_{G_n} \int_{H_n} \Phi(\langle U_g U_h \Omega_{n,\epsilon}, \rho U_g U_h \Omega_{n,\epsilon} \rangle) dh dg.$$

These are integrals of continuous bounded functions on compact groups.

For fixed g , identify the positive compression $P U_g^* \rho U_g P$ with $|1\rangle\langle 1| \otimes A_g$. Its trace is $a_\rho(g)$. If this trace is positive, define $\rho_g = A_g / a_\rho(g)$ on $\mathcal{F}_{n-1}^{\epsilon-1}$. The measurement value in the inner integral is exactly

$$(3.12) \quad a_\rho(g) \langle U_h \Omega_{n-1,\epsilon-1}, \rho_g U_h \Omega_{n-1,\epsilon-1} \rangle.$$

If $a_\rho(g) = 0$, positivity and trace zero give $A_g = 0$, and the inner integral is $\Phi(0)$. In both cases the outer measure remains the original Haar probability dg .

Let R have the Husimi distribution of a pure Gaussian state in the smaller sector, and put $H_\Phi(a) = \mathbb{E}\Phi(aR)$. This function is continuous and convex on $[0, 1]$. The induction hypothesis, applied for each fixed $a = a_\rho(g)$, and Lemma 3.2 give

$$(3.13) \quad \begin{aligned} \int_{G_n} \Phi(Q_\rho) dg &\leq \int_{G_n} H_\Phi(a_\rho(g)) dg \\ &\leq \sum_s t_s \int_{G_n} H_\Phi(a_{\chi_s}(g)) dg \\ &= \int_{G_n} H_\Phi(a_{\Omega_{n,\epsilon}}(g)) dg = \int_{G_n} \Phi(Q_{|\Omega_{n,\epsilon}\rangle\langle\Omega_{n,\epsilon}|}) dg. \end{aligned}$$

The middle inequality is the common-mixture Jensen step of Lemma 2.2, with probabilities fixed for all g . The next equality follows from Haar invariance of the Gaussian orbit. The last follows from Lemma 3.3: every nonzero conditional state of the Gaussian input is a lower-dimensional Gaussian state, so the inner integral has the same law at each g . Zero branches give $\Phi(0)$ on both sides.

Suppose now that Φ is strictly convex. By (3.4), $\mathbb{E}R = 2^{-(n-2)} > 0$, and hence H_Φ is strictly convex. Consider two positive-weight endpoints of the covariance mixture with difference $\Delta \neq 0$. Choose a real unit u with $\Delta u \neq 0$ and set $v = -\Delta u / \|\Delta u\|$. Skew-symmetry implies that u, v are orthonormal and $u^\top \Delta v \neq 0$. For $n \geq 2$, $\text{SO}(2n)$ is transitive on ordered orthonormal two-frames: complete either frame to an oriented orthonormal basis and map one basis to the other. Thus some g distinguishes the two endpoint probabilities in (3.7). Continuity preserves this difference on a nonempty open set, which has positive Haar measure. Jensen's inequality in (3.13) is then strict.

Equality therefore forces all positive-weight endpoints in the construction of Lemma 3.2 to have the same covariance. Distinct occupation strings have distinct diagonal covariances, so exactly one t_s is nonzero. For a positive operator,

$$|\langle s, \sigma t \rangle|^2 \leq \langle s, \sigma s \rangle \langle t, \sigma t \rangle;$$

this is Cauchy-Schwarz applied to $\sigma^{1/2}|s\rangle$ and $\sigma^{1/2}|t\rangle$. The unique nonzero diagonal entry, together with trace one, forces $\sigma = |s\rangle\langle s|$. Undoing the Spin rotation shows that ρ is a pure Gaussian projector. Such projectors attain equality by Haar invariance. $\square$

Example 3.4 (Occupation conditioning and shared probabilities). In two modes, $\chi_\theta = \cos \theta |00\rangle + \sin \theta |11\rangle$ is Gaussian: it is obtained from the vacuum by $\exp[\theta(a_1^* a_2^* - a_2 a_1)]$, whose generator equals $(c_1 c_3 - c_2 c_4)/2$. Its first-mode occupation probability is $\sin^2 \theta$; whenever this is nonzero, the remaining conditional state is the one-mode occupied ray.

In four modes, consider $\psi = (|0000\rangle + |1111\rangle)/\sqrt{2}$. Its covariance is zero: each mode is occupied with probability $1/2$, and a quadratic Majorana expression cannot connect its two occupation components. Thus its covariance mixture uses the two fixed weights $1/2, 1/2$ on the vacuum and the fully occupied state. For every rotated mode, (3.8) gives $a_\psi(g) = 1/2$. The state is non-Gaussian, since a Gaussian covariance is an orthogonal conjugate of the vacuum covariance and therefore squares to $-\text{Id}$. Its first-mode occupied conditional state is nevertheless the Gaussian vector $|111\rangle$. This example separates covariance mixing, conditional states, and the original many-body state.

3.5. The basic odd-spin representation. The odd-dimensional spin representation has the same coherent rays as a half-spin representation in one larger real dimension. Equality of the invariant probability measures on those rays transfers the distribution comparison without altering its normalization.

Corollary 3.5 (Odd-spin Husimi convex order). *For every $m \geq 1$, the basic 2^m -dimensional representation of $\text{Spin}(2m+1)$ satisfies the conclusion of Theorem 3.1 on its highest-weight coherent orbit. For a strictly convex test function, equality holds exactly at the pure coherent projectors of that orbit.*

Proof. Set $N = m + 1$. Fix a real unit vector $e \in \mathbb{R}^{2N}$ and embed $H = \text{Spin}(2N - 1)$ as the lift of its stabilizer in $\text{SO}(2N)$. The restriction of either half-spin representation to H is the basic spin representation. One algebraic verification uses

$$\text{Cl}_{2N-1}^0(\mathbb{C}) \simeq \text{Cl}_{2N-2}(\mathbb{C}) \simeq \text{Mat}_{2^{N-1}}(\mathbb{C}).$$

The induced unital action on a half-spin space of dimension 2^{N-1} is the standard irreducible matrix-module action. The Clifford identities follow by using products with one fixed unit generator for the first isomorphism and the creation and annihilation matrix units for the second. Under a maximal torus of H , the weights are $(\pm\frac{1}{2}, \dots, \pm\frac{1}{2})$, each with multiplicity one: they are the restrictions of the half-spin weights, with the last sign determined by parity. In particular, a highest-weight ray is an occupation Gaussian ray.

For completeness we identify the entire orbit. The proof of Lemma 3.3 identifies maximal isotropic annihilators with orthogonal complex structures on $\mathbb{R}^{2N}$. For example, with $z_j = (p_j + iq_j)/\sqrt{2}$ one may take $Jp_j = q_j$ and $Jq_j = -p_j$. This construction is independent of the unitary basis of the isotropic space, and the two parity orbits correspond to its two possible orientation components.

Let J_1, J_2 lie in the same component. An element of $\text{SO}(2N - 1)$ fixing e can first send $J_1 e$ to $J_2 e$. After this conjugation the two complex structures agree on $\text{Span}(e, J_2 e)$. On its orthogonal complement they induce orthogonal complex structures of the same orientation. Choose adapted oriented orthonormal bases there; the map between these bases belongs to $\text{SO}(2N - 2)$ and conjugates one restricted complex structure to the other. Extending it by the identity on $\text{Span}(e, J_2 e)$ fixes e and completes the conjugation. Thus H is transitive on the entire half-spin coherent orbit.

The pushforwards of Haar probability from $\text{Spin}(2N)$ and from H coincide: both are invariant probabilities on this compact transitive H -space. Uniqueness follows by averaging any continuous function over H ; the average is constant on the orbit, so every invariant probability has the same integral. The Husimi functions are those of the same rays in the same Hilbert space. Theorem 3.1, with $n = N$, therefore gives the asserted inequality and equality criterion. $\square$

4. COHERENT DISTRIBUTIONS AND ENTROPY MINIMA

4.1. Normalization and entropy conventions. In this section, $(\mathcal{O}, \nu)$ denotes either the Slater orbit with its invariant probability measure from Section 2, or a fixed-parity Gaussian orbit from Theorem 3.1. The representation dimension is respectively $D = \binom{n}{p}$ or $D = 2^{n-1}$. The same

definitions apply to the odd-spin orbit in Corollary 3.5. A unit coherent vector at $x \in \mathcal{O}$ is denoted by Ω_x , and $Q_\rho(x) = \langle \Omega_x, \rho \Omega_x \rangle$ is independent of its phase.

The resolution identity gives

$$(4.1) \quad \int_{\mathcal{O}} |\Omega_x\rangle \langle \Omega_x| d\nu(x) = \frac{\text{Id}}{D}, \quad D \int_{\mathcal{O}} Q_\rho d\nu = 1.$$

For exterior powers this is (2.2). For the irreducible spin representations, the averaged projector commutes with the group; Schur's lemma and its trace one give the identity.

Define the unnormalized and normalized positive moments by

$$(4.2) \quad M_\rho(q) = \int_{\mathcal{O}} Q_\rho^q d\nu, \quad N_\rho(q) = D M_\rho(q), \quad q > 0.$$

The normalized first moment is one. Since $Q_\rho \geq 0$ has positive integral, each $M_\rho(q)$ is strictly positive. All these moments are finite because $0 \leq Q_\rho \leq 1$.

We use the resolution measure $D d\nu$ to define the Rényi–Wehrl and Wehrl entropies:

$$(4.3) \quad S_q(\rho) = \frac{\log N_\rho(q)}{1 - q}, \quad q > 0, \quad q \neq 1,$$

$$(4.4) \quad S_W(\rho) = -D \int_{\mathcal{O}} Q_\rho \log Q_\rho d\nu.$$

Logarithms are natural and $0 \log 0 = 0$. In particular, the latter integrand is the continuous extension of $x \log x$ at zero. The probability density relative to ν is $f_\rho = D Q_\rho$; its differential entropy is

$$(4.5) \quad - \int f_\rho \log f_\rho d\nu = S_W(\rho) - \log D.$$

4.2. The coherent Beta laws. For $a, b > 0$, the distribution $\text{Beta}(a, b)$ on $(0, 1)$ has density

$$\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} t^{a-1} (1-t)^{b-1},$$

where $\Gamma(z) = \int_0^\infty s^{z-1} e^{-s} ds$ for $z > 0$ is Euler's Gamma function. A $\text{Gamma}(a, 1)$ variable has density $s^{a-1} e^{-s} / \Gamma(a)$ on $(0, \infty)$. If X and Y are independent Gamma variables with shapes a, b and common scale one, the change of variables $(X, Y) = (rt, r(1-t))$ shows that $X/(X+Y)$ has distribution $\text{Beta}(a, b)$.

Proposition 4.1 (Coherent measurement laws). *For a Slater input in $\mathcal{H}_{p,n}$ with $0 < p < n$,*

$$(4.6) \quad Q_{\rho_W} \stackrel{\text{law}}{=} \prod_{j=0}^{p-1} A_j, \quad A_j \sim \text{Beta}(p-j, n-p),$$

where the A_j are independent. For $p = 0$ or $p = n$, the coherent measurement is identically one.

For a pure Gaussian input in either parity sector $\mathcal{F}_n^\epsilon$,

$$(4.7) \quad Q_\Omega \stackrel{\text{law}}{=} \prod_{k=1}^{n-1} B_k, \quad B_k \sim \text{Beta}(k, k),$$

with independent factors. The product is one when $n = 1$. For the basic spin representation of $\text{Spin}(2m+1)$, use (4.7) with $n = m+1$.

Proof. Let Z be a standard complex Gaussian vector in $E = \mathbb{C}^n$, with independent coordinates of density $\pi^{-1} e^{-|z|^2}$. Its direction $u = Z/\|Z\|$ is uniform on the complex unit sphere. For a fixed p -plane W , the variables

$$X = \|\pi_W Z\|^2, \quad Y = \|(\text{Id} - \pi_W)Z\|^2$$

are independent Gamma variables of shapes p and $n - p$. Consequently the first conditional weight is

$$(4.8) \quad a_{\rho_W}(u) = \|\pi_W u\|^2 = \frac{X}{X + Y} \sim \text{Beta}(p, n - p).$$

For almost every u this weight is positive. By Lemma 2.5, the normalized conditional input is a Slater density in $\Lambda^{p-1}(u^\perp)$. Its inner Haar Husimi distribution depends only on $(p - 1, n - 1)$, independently of u . On the double-Haar probability space of (2.14), this conditional distribution being constant means that the lower-dimensional readout is independent of the first weight. Iterating (2.13) proves (4.6); at step j the two Beta shapes are $p - j$ and $(n - j) - (p - j) = n - p$.

For a Gaussian input, its Majorana covariance Γ_Ω satisfies $\Gamma_\Omega^2 = -\text{Id}$. A Haar-rotated first mode gives a uniform oriented orthonormal two-frame (u, v) in $\mathbb{R}^{2n}$. Conditioned on u , the vector v is uniform on the unit sphere in $u^\perp$. The linear functional $v \mapsto u^\top \Gamma_\Omega v$ is represented there by the unit vector $\Gamma_\Omega^\top u$. For $n \geq 2$, its value t therefore has density proportional to

$$(1 - t^2)^{n-2}, \quad -1 < t < 1.$$

The occupation identity (3.7) gives $a = (1 + t)/2$, whose density is proportional to $a^{n-2}(1 - a)^{n-2}$. Thus $a \sim \text{Beta}(n - 1, n - 1)$. The Gaussian conditional closure in Lemma 3.3 and the double-Haar identity (3.11) make the remaining coherent readout independent of this weight, with the law for $n - 1$ modes. Induction proves (4.7). The one-dimensional initial sector gives the empty product. Finally, Corollary 3.5 identifies the entire odd-spin coherent orbit and its probability measure with the indicated half-spin orbit. $\square$

The independence in Proposition 4.1 describes coherent inputs. For a general density, its normalized conditional state varies with the preceding measurement direction; the convex-order proof retains that dependence.

4.3. Positive moments and explicit minima. The Beta integral gives

$$(4.9) \quad \mathbb{E}A^q = \frac{\Gamma(a + q)\Gamma(a + b)}{\Gamma(a)\Gamma(a + b + q)}, \quad A \sim \text{Beta}(a, b), \quad q > 0.$$

Consequently the coherent moments in the two series are

$$(4.10) \quad m_{p,n}(q) := \prod_{j=0}^{p-1} \frac{\Gamma(p - j + q)\Gamma(n - j)}{\Gamma(p - j)\Gamma(n - j + q)},$$

$$(4.11) \quad b_n(q) := \prod_{k=1}^{n-1} \frac{\Gamma(k + q)\Gamma(2k)}{\Gamma(k)\Gamma(2k + q)}.$$

Here $m_{0,n}(q) = 1$ by the empty-product convention, while each factor in $m_{n,n}(q)$ cancels to one. These formulas cover every real $q > 0$. In particular,

$$(4.12) \quad m_{p,n}(1) = \binom{n}{p}^{-1}, \quad b_n(1) = 2^{-(n-1)},$$

as required by (4.1).

Theorem 4.2 (Wehrl and Rényi–Wehrl minima). *Let $H_0 = 0$ and $H_r = \sum_{j=1}^r j^{-1}$ for $r \geq 1$. For every density on $\mathcal{H}_{p,n}$, $0 \leq p \leq n$,*

$$(4.13) \quad S_q(\rho) \geq \frac{\log \left[\binom{n}{p} m_{p,n}(q) \right]}{1 - q}, \quad q > 0, \quad q \neq 1,$$

$$(4.14) \quad S_W(\rho) \geq s_{p,n} := \sum_{j=0}^{p-1} (H_{n-j} - H_{p-j}).$$

Equality in each inequality holds precisely for pure Slater densities.

For every density supported on the fixed parity sector $\mathcal{F}_n^\epsilon$, $n \geq 1$,

$$(4.15) \quad S_q(\rho) \geq \frac{\log[2^{n-1} b_n(q)]}{1 - q}, \quad q > 0, \quad q \neq 1,$$

$$(4.16) \quad S_W(\rho) \geq s_n^{\text{spin}} := \sum_{k=1}^{n-1} (H_{2k} - H_k).$$

Equality holds precisely for pure Gaussian densities in that parity sector. For the basic spin representation of $\text{Spin}(2m+1)$, the same formulas hold with dimension 2^m , moment $b_{m+1}(q)$, and constant s_{m+1}^{spin} .

Proof. For $q > 1$, the function $x \mapsto x^q$ is continuous and strictly convex on $[0, 1]$. Theorems 2.1 and 3.1 imply that its moment is maximized by a pure coherent input. Multiplication by D , application of the increasing logarithm, and division by $1 - q < 0$ give the entropy lower bound. For $0 < q < 1$, use the continuous strictly convex function $x \mapsto -x^q$. The moment inequality reverses, and $1 - q > 0$, giving the same direction for the entropy inequality. The positivity of the moments justifies every logarithm and its equality criterion. Strict convexity identifies the equality cases in both ranges.

For Wehrl entropy, apply the same theorems to the continuous strictly convex function $x \mapsto x \log x$, with its value zero at the origin, and multiply by $-D$. It remains to evaluate the coherent integral. Differentiation of a moment at $q = 1$ gives

$$\left. \frac{d}{dq} \int Q_\Omega^q d\nu \right|_{q=1} = \int Q_\Omega \log Q_\Omega d\nu.$$

This differentiation is justified directly: for $q \in [1/2, 3/2]$ and $0 < x \leq 1$, $|x^q \log x| \leq x^{1/2} |\log x| \leq 2/e$. The derivative at $x = 0$ is zero, so dominated convergence applies on the probability space including all zeros of Q_Ω .

Taking logarithmic derivatives in (4.10) and (4.11), and using $\Gamma(z+1) = z\Gamma(z)$, gives

$$\begin{aligned} -D_{p,n} m'_{p,n}(1) &= \sum_{j=0}^{p-1} (H_{n-j} - H_{p-j}), \\ -2^{n-1} b'_n(1) &= \sum_{k=1}^{n-1} (H_{2k} - H_k). \end{aligned}$$

For clarity, if $\psi_\Gamma = \Gamma'/\Gamma$, the recurrence $\psi_\Gamma(r+1) = \psi_\Gamma(r) + 1/r$ shows that each difference $\psi_\Gamma(b+1) - \psi_\Gamma(a+1)$ at positive integers equals $H_b - H_a$. Equation (4.12) supplies the factor D in the preceding derivatives. This proves the displayed constants. The odd-spin conclusions follow from Corollary 3.5. $\square$

Example 4.3 (Small sectors). For one particle in n modes, the coherent Wehrl value is $s_{1,n} = H_n - 1$. For two particles in four modes,

$$s_{2,4} = (H_4 - H_2) + (H_3 - H_1) = \frac{17}{12}.$$

The maximally mixed density on this six-dimensional sector has $Q \equiv 1/6$ and hence $S_W = \log 6$, larger than the coherent value. In the spin series the constants for $n = 1, 2, 3$ are 0, $1/2$, and $13/12$. The empty and filled exterior sectors have zero Wehrl and Rényi–Wehrl entropy.

All formulas above concern the basic exterior, half-spin, and odd-spin representations specified in the corresponding theorems. The representation parameter is not replaced by a higher Cartan power.

5. A POLYNOMIAL OBSTRUCTION TO THREE GAUSSIAN SUMMANDS

A Gaussian decomposition is an exact vector identity. Its coefficients may have arbitrary complex phases, and its summands may pair modes belonging to different prescribed blocks. Consequently, estimates on each summand in isolation do not determine the minimum number of summands. We use a polynomial of one complete Clifford Gram operator: a common change of coordinates transports every summand and every Gram entry simultaneously.

Let $\mathcal{F}_n = \Lambda\mathbb{C}^n$ and $\mathcal{F}_n^+ = \Lambda^{\text{even}}\mathbb{C}^n$. Products of vectors in an exterior algebra below mean wedge products. For an ordered direct sum $E_A \oplus E_B$, we identify $\Lambda E_A \otimes \Lambda E_B$ with $\Lambda(E_A \oplus E_B)$ by $s \otimes t \mapsto s \wedge t$, with all A modes before the B modes. Denote by $\hat{\mathcal{G}}_n^+$ the nonzero complex cone over the vacuum orbit of the compact Euclidean group $\text{Spin}(2n)$. For $0 \neq \psi \in \mathcal{F}_n^+$ set

$$(5.1) \quad \chi_G(\psi) = \min \left\{ r : \psi = g_1 + \cdots + g_r, \quad g_j \in \hat{\mathcal{G}}_n^+ \right\}.$$

Every even occupation vector belongs to this cone, so the minimum exists. Allowing odd Gaussian summands gives the same minimum for an even target: the sum of all odd summands is zero, and deleting them preserves the vector without increasing the number of terms. Nonzero scalar multiplication also preserves (5.1), in both directions.

Theorem 5.1. *For any nonzero $\psi, \phi \in \mathcal{F}_4^+$,*

$$(5.2) \quad \chi_G(\psi \otimes \phi) = \chi_G(\psi) \chi_G(\phi).$$

Each factor has rank one or two. If both factors are non-Gaussian, their product has rank four. In particular, for

$$M = \frac{|0000\rangle + |1111\rangle}{\sqrt{2}}, \quad \chi_G(M^{\otimes 2}) = 4.$$

The decomposition on the left permits arbitrary eight-mode Gaussian terms, including terms with pairing across the two four-mode blocks.

The last assertion is the unrestricted two-copy rank question posed by Cudby–Strelchuk [5]. Gaussian fidelity and extent, for which four-mode multiplicativity results were obtained in [6, 7], are different quantities: they retain normalization or coefficient costs, whereas (5.1) counts nonzero terms in an exact sum.

5.1. Clifford conventions and the physical cone. Write $E = \mathbb{C}^n$, $U = E^* \oplus E$, and

$$(5.3) \quad c(f, v) = \iota_f + \varepsilon_v, \quad \varepsilon_v s = v \wedge s, \quad q(f, v) = f(v).$$

The polar form is $b((f, v), (g, w)) = f(w) + g(v)$, and

$$(5.4) \quad c(u)^2 = q(u)\text{Id}, \quad c(u)c(w) + c(w)c(u) = b(u, w)\text{Id}.$$

A nonzero $s \in \mathcal{F}_n$ is pure if $\text{Ann}(s) = \{u : c(u)s = 0\}$ has dimension n . The CAR make this annihilator isotropic, so its dimension is at most n . The common kernel of all contractions is the

scalar vacuum line. Indeed, if $s = \sum_I s_I e_I$ has a nonempty coefficient s_I , contracting by any $i \in I$ reads s_I , up to sign, in the $I \setminus \{i\}$ coefficient; no other source coefficient contributes to that same output.

By Lemma 3.3, even pure-spinor rays are exactly the compact Gaussian vacuum rays; see also [9]. To specify the real form in the split coordinates, use the standard Hermitian basis of E and define

$$\overline{(f, v)} = (v^\flat, f^\sharp), \quad v^\flat(e_i) = \overline{v_i}, \quad f^\sharp = \sum_i \overline{f(e_i)} e_i.$$

This exchanges the two split summands and is not coordinatewise conjugation on $E^* \oplus E$. It satisfies $b(u, \bar{u}) > 0$ for $u \neq 0$. A maximal isotropic subspace L therefore has $L \cap \bar{L} = 0$. An orthonormal basis of L gives, by taking real and imaginary parts, a real orthogonal frame, and an orthogonal map sends the vacuum annihilator to L . A lift to the Pin group transports the corresponding annihilator. Its spinor line is unique, because transport back to the vacuum reduces uniqueness to the common contraction kernel. If that line is even, the lift has even Clifford parity and lies in Spin. Thus the compact orbit gives every even pure ray. The converse follows from the CAR transport of the annihilator. In particular,

$$(5.5) \quad \hat{\mathcal{G}}_n^+ = \{s \in \mathcal{F}_n^+ : s \neq 0, \dim \text{Ann}(s) = n\}.$$

Consequently complex Spin transformations preserve the same physical cone of rays. They are used below as invertible coordinate transformations, not as norm-preserving physical protocols.

For later use, a nonzero isotropic Clifford multiplication preserves purity. If s is pure, $q(u) = 0$, and $c(u)s \neq 0$, then $u \notin \text{Ann}(s)$ and

$$\mathbb{C}u + (\text{Ann}(s) \cap u^\perp) \subseteq \text{Ann}(c(u)s).$$

The left side has dimension n : the restriction of $b(u, \cdot)$ to the maximal isotropic space $\text{Ann}(s)$ is nonzero, and u does not belong to that space. The upper dimension bound therefore gives purity. Two such multiplications prove the closure of a nonzero occupation or vacancy projection, consistent with Lemma 3.3.

5.2. The whole vacuum chart from elementary units. Fix an ordered basis $e_0, \dots, e_{n-1}$ and dual basis $f_0, \dots, f_{n-1}$. Write $\varepsilon_i = \varepsilon_{e_i}$ and $\iota_i = \iota_{f_i}$. Let $[s]_0$ denote the vacuum coefficient and $[s]_2 \in \Lambda^2 E$ the degree-two homogeneous component. Suppose s is pure and $a = [s]_0 \neq 0$. Projection of $\text{Ann}(s)$ to E^* is injective: if $\varepsilon_v s = 0$, its degree-one part is $av = 0$. Since both spaces have dimension n , the projection is an isomorphism. It defines a linear map $\mathcal{D} : E^* \rightarrow E$ with

$$(5.6) \quad (\iota_f + \varepsilon_{\mathcal{D}f})s = 0 \quad (f \in E^*), \quad f(\mathcal{D}g) + g(\mathcal{D}f) = 0.$$

The second identity follows from the CAR. Applying a second contraction and then reading degree zero gives

$$(5.7) \quad g(\mathcal{D}f) = -\frac{[\iota_g \iota_f s]_0}{a}.$$

These are equations on the complete original vector, not just its first two homogeneous components.

Lemma 5.2 (Elementary chart reconstruction). *For $i < j$ define*

$$a_{ij} = f_i(\mathcal{D}f_j) = -f_j(\mathcal{D}f_i), \quad \mathcal{U}(\mathbf{a}) = \prod_{i < j} (\text{Id} + a_{ij} \varepsilon_i \varepsilon_j), \quad \Omega(\mathbf{a}) = \prod_{i < j} (1 + a_{ij} e_i e_j).$$

Here $\mathbf{a} = (a_{ij})_{i < j}$ is the coefficient array, distinct from the vacuum coefficient a . The factors may be taken in any fixed order. Then $s = a\Omega(\mathbf{a})$. More generally, vectors annihilated by the same graph $\iota_f + \varepsilon_{\mathcal{D}f}$ and having the same vacuum coefficient are equal.

Proof. Put $N_{ij} = \varepsilon_i \varepsilon_j$. The creation CAR give $N_{ij}^2 = 0$, $[N_{ij}, \varepsilon_v] = 0$, and $[N_{ij}, N_{kl}] = 0$, including repeated-index cases. Thus $U_{ij}(z) = \text{Id} + zN_{ij}$ has inverse $\text{Id} - zN_{ij}$. For $h_{ij}(f) = f(e_i)e_j - f(e_j)e_i$, the mixed CAR give

$$[\iota_f, N_{ij}] = \varepsilon_{h_{ij}(f)}, \quad N_{ij}\varepsilon_{h_{ij}(f)} = \varepsilon_{h_{ij}(f)}N_{ij} = 0.$$

It follows, on all of $\mathcal{F}_n$, that

$$(5.8) \quad U_{ij}(z)^{-1} \iota_f U_{ij}(z) = \iota_f + z\varepsilon_{h_{ij}(f)}.$$

Set $h_{\mathbf{a}}(f) = \sum_{i < j} a_{ij} h_{ij}(f)$. Evaluating on the dual basis and using (5.6), including its zero diagonal, gives $h_{\mathbf{a}} = -\mathcal{D}$. All factors commute with every creator, so composing (5.8) gives

$$\iota_f \mathcal{U}(\mathbf{a})^{-1} s = \mathcal{U}(\mathbf{a})^{-1} (\iota_f + \varepsilon_{\mathcal{D}f}) s = 0.$$

Each factor and its inverse preserve degree zero. The common contraction kernel is the vacuum line, hence $\mathcal{U}(\mathbf{a})^{-1} s = a \cdot 1$, and multiplication by $\mathcal{U}(\mathbf{a})$ proves the assertion. Applying the same inverse product to two vectors proves uniqueness. No expansion of their higher-degree components is needed. $\square$

For reference, if $\omega = \sum_{i < j} a_{ij} e_i e_j$, then $\Omega(\mathbf{a}) = \exp(\omega)$: the degree-two monomials commute and square to zero, so their finite exponentials multiply to the displayed product. The proof of Lemma 5.2 does not require a general exponential construction.

For $g \in \text{GL}(E)$ the original exterior map satisfies

$$(5.9) \quad \iota_f \Lambda g = \Lambda g \iota_{f \circ g}, \quad \varepsilon_{gv} \Lambda g = \Lambda g \varepsilon_v.$$

The first identity follows by induction on exterior monomials from the contraction rule, and the second from multiplicativity. Write $\Omega_{\mathcal{D}}$ for the chart product with coefficients $a_{ij} = f_i(\mathcal{D}f_j)$, and let $g^\vee : E^* \rightarrow E^*$ denote the algebraic dual map $g^\vee f = f \circ g$, with no complex conjugation. The new graph is

$$(5.10) \quad \mathcal{D}^g(f) = g(\mathcal{D}(f \circ g)).$$

Both $\Lambda g \Omega_{\mathcal{D}}$ and $\Omega_{\mathcal{D}^g}$ have this graph and vacuum coefficient one. Lemma 5.2 therefore proves their equality. In particular a nondegenerate alternating graph can be reduced to its standard form by a single g acting on the whole Fock space.

Here is an explicit linear-algebra justification of that reduction. For a nondegenerate alternating form choose u, v with pairing one. Their span is nondegenerate. Subtracting the appropriate multiples of u, v from any vector makes it orthogonal to both, giving the direct sum of this plane and its orthogonal complement. The complement is nondegenerate, since a vector orthogonal to both summands is orthogonal to the whole space. Repeating produces a symplectic basis. Applied to the alternating tensor $\mathcal{D} : E^* \rightarrow E$, this is exactly the congruence $\mathcal{D} \mapsto g\mathcal{D}g^\vee$ in (5.10).

Example 5.3. The four-mode vector $1 + \kappa e_0 e_1 e_2 e_3$, with $\kappa \neq 0$, is not Gaussian. Its degree-two component vanishes, so (5.7) would force $\mathcal{D} = 0$ if it were pure; Lemma 5.2 would then make it a vacuum multiple. It is a sum of two Gaussian occupation vectors, and hence has rank two. Its two-copy expansion has four terms, but this observation alone does not rule out a shorter expansion using eight-mode Gaussians.

5.3. A relative invariant of the complete Clifford Gram. For the remainder of the rank-four proof let $n = 8$ and $\text{vol} = e_0 \cdots e_7$. Fix the bilinear Chevalley convention

$$(5.11) \quad \mathcal{B}(s, t) = [s \wedge \text{rev}(t)]_{\text{vol}}, \quad \text{rev} |_{\wedge^j E} = (-1)^{j(j-1)/2} \text{Id}.$$

Reversal acts on the second leg; no complex conjugation is present. For an ordered subset $I \subseteq \{0, \dots, 7\}$,

$$(5.12) \quad \mathcal{B}(e_I, e_{I^c}) = (-1)^{\sum_{i \in I} (i+1)}.$$

Indeed, for $|I| = k$ the wedge-order sign has exponent $\sum_{i \in I} i - k(k-1)/2$, and reversing the second leg adds $(8-k)(7-k)/2$. Their sum is congruent to $\sum_{i \in I} (i+1)$ modulo two. This calculation applies in every degree. The form is symmetric and nondegenerate.

For every Clifford direction u , the exterior multiplication and contraction rules give

$$(5.13) \quad \mathcal{B}(c(u)s, t) = -\mathcal{B}(s, c(u)t).$$

For creation, move the created vector past the other seven factors in a nonzero top-degree term; for contraction, apply contraction to the identically zero degree-nine exterior product. Both yield the displayed minus sign with (5.11). Consequently an elementary creation unit, or its two-contraction counterpart, preserves $\mathcal{B}$: its generator is skew-adjoint for $\mathcal{B}$ and squares to zero. Also $\tau_i = \varepsilon_i + \iota_{f_i}$ has $\tau_i^2 = \text{Id}$ and multiplier -1 , so an even product of the τ_i preserves $\mathcal{B}$. Finally,

$$(5.14) \quad \mathcal{B}(\Lambda g s, \Lambda g t) = \det(g)\mathcal{B}(s, t).$$

This follows by applying Λg to the top exterior product.

Let $W = \Lambda^2 U$ and equip it with

$$G(u \wedge v, w \wedge z) = b(u, w)b(v, z) - b(u, z)b(v, w).$$

Define the whole bivector source map and its raised Gram by

$$(5.15) \quad J_\psi(u \wedge v) = \frac{1}{2}[c(u), c(v)]\psi, \quad H_\psi(\xi, \zeta) = \mathcal{B}(J_\psi \xi, J_\psi \zeta), \quad A_\psi = G^{-1}H_\psi.$$

All $\binom{16}{2} = 120$ bivector directions occur here. The matrix of A_ψ is $G^{-1}J_\psi^\top \mathcal{B} J_\psi$; it need not be Hermitian or positive. Set

$$(5.16) \quad s_\psi = \frac{1}{2}\mathcal{B}(\psi, \psi), \quad t_k(\psi) = \text{Tr}(A_\psi^k),$$

$$(5.17) \quad F_{16}(\psi) = 576s_\psi^2 t_6(\psi) - 7t_4(\psi)^2 - 18960s_\psi^4 t_4(\psi) + 4205376s_\psi^8, \quad P_{18}(\psi) = s_\psi F_{16}(\psi).$$

These are homogeneous polynomials of degrees 16 and 18 in ψ .

Lemma 5.4. *Suppose invertible maps T on $\mathcal{F}_8$ and R on U satisfy*

$$Tc(u)T^{-1} = c(Ru), \quad R^*b = b, \quad \mathcal{B}(Ts, Tt) = \lambda \mathcal{B}(s, t), \quad \lambda \neq 0.$$

With $L = \Lambda^2 R$, one has

$$(5.18) \quad A_{T\psi} = \lambda L A_\psi L^{-1}, \quad F_{16}(T\psi) = \lambda^8 F_{16}(\psi), \quad P_{18}(T\psi) = \lambda^9 P_{18}(\psi).$$

The elementary units above, even particle-hole products, and Λg all satisfy these hypotheses, with their stated form multipliers.

Proof. The complete source map obeys $J_{T\psi} = T J_\psi L^{-1}$. The transformation of H follows by applying the same form identity to both of its arguments. Since L preserves G , raising an index gives the first formula. Hence $s_{T\psi} = \lambda s_\psi$ and $t_k(T\psi) = \lambda^k t_k(\psi)$; substitution gives the other formulas. For elementary units the Clifford conjugation is precisely the shear (5.8), whose graph is alternating, so it preserves q and b . Two-contraction units are dual. The CAR give the orthogonal reflection, up to an overall minus sign, for τ_i . Equation (5.9) gives the direction change $(f, v) \mapsto (f \circ g^{-1}, gv)$ for Λg . The form multipliers were established in (5.13)–(5.14). $\square$

5.4. The two complete coordinate evaluations. We give the coordinates explicitly so that the finite identities below can be checked independently of a spectral or invariant-theoretic classification. Write $u_i^+ = (0, e_i)$ and $u_i^- = (f_i, 0)$ in U , and order the sixteen directions as $z_0, \dots, z_{15} = (u_0^+, \dots, u_7^+, u_0^-, \dots, u_7^-)$. An integer $0 \leq m < 256$ encodes an occupied subset through its bits m_i ; e_m is the increasing exterior product on that subset. Put

$$\kappa(m) = (-1)^{\sum_{i=0}^7 (i+1)m_i}, \quad X_i(m) = m \text{ xor } 2^{i \bmod 8}.$$

If $i < 8$, let $\gamma_i(m) = (-1)^{\sum_{r < i} m_r}$ when $m_i = 0$ and zero otherwise. If $i \geq 8$, use the same sign with i replaced by $i - 8$, and require $m_{i-8} = 1$. Thus $c(z_i)e_m = \gamma_i(m)e_{X_i(m)}$. For a sorted pair $\alpha = (i, j)$ set

$$X_\alpha = X_i X_j, \quad j_\alpha(m) = 2\gamma_j(m)\gamma_i(X_j(m)) - \mathbf{1}_{j=i+8}.$$

Then $2J_{e_m}(\alpha) = j_\alpha(m)e_{X_\alpha(m)}$. Replace each direction in α by its dual, reorder increasingly, and call the resulting pair $\alpha^\vee$ and the reordering sign ϵ_α . The inverse of G is exactly this signed dual-pair operation. Therefore, for $\psi = \sum_{m=0}^{255} \psi_m e_m$,

$$(5.19) \quad (A_\psi)_{\alpha\beta} = \frac{\epsilon_\alpha}{4} \sum_{m=0}^{255} \psi_m j_{\alpha^\vee}(m) \kappa(X_{\alpha^\vee}(m)) j_\beta(v) \psi_v, \quad v = X_\beta(255 \text{ xor } X_{\alpha^\vee}(m)).$$

This is just (5.15) expanded in the original occupation basis. The factor $1/4$ accounts for the two occurrences of $2J$. Every encoded subset and every direction in this formula stays in the original finite domain.

Lemma 5.5. *Let*

$$\Omega_* = (1 + e_0 e_1)(1 + e_2 e_3)(1 + e_4 e_5)(1 + e_6 e_7), \quad \eta = \alpha \mathbf{1} + \beta \text{vol} + \gamma \Omega_*.$$

For arbitrary $\alpha, \beta, \gamma \in \mathbb{C}$, set $x = \alpha\beta$, $y = \alpha\gamma$, $z = \beta\gamma$, $s = x + y + z$, and $w = xyz$. Then

$$(5.20) \quad s_\eta = s,$$

$$(5.21) \quad t_4(\eta) = 312s^4 - 1728sw,$$

$$(5.22) \quad t_6(\eta) = 4152s^6 - 69984s^3w + 36288w^2.$$

In particular $F_{16}(\eta) = P_{18}(\eta) = 0$.

Proof. Here is a complete common basis for the matrix calculation. Partition the mode indices into $B_r = \{2r, 2r+1\}$, $0 \leq r < 4$, and put $\bar{u} = u \text{ xor } 1$, $\sigma_u = (-1)^u$. There are eight zero directions

$$u_{2r}^+ \wedge u_{2r+1}^-, \quad u_{2r+1}^+ \wedge u_{2r}^- \quad (0 \leq r < 4).$$

For each $r < s$, $i \in B_r$, $j \in B_s$, take the signed quartet

$$(5.23) \quad (-u_i^+ \wedge u_j^+, -\sigma_i u_j^+ \wedge u_i^-, \sigma_j u_i^+ \wedge u_j^-, -\sigma_i \sigma_j u_i^- \wedge u_j^-).$$

These give 24 four-dimensional blocks. The remaining sixteen directions are, for each r , the four entries

$$(5.24) \quad (u_{2r}^+ \wedge u_{2r+1}^+, \quad u_{2r}^+ \wedge u_{2r}^-, \quad u_{2r+1}^+ \wedge u_{2r+1}^-, \quad u_{2r}^- \wedge u_{2r+1}^-).$$

This list contains every pair of the sixteen split directions exactly once. Substitution in (5.19) gives the following whole matrix:

$$(5.25) \quad A_\eta \simeq 0_8 \oplus T^{\oplus 24} \oplus (T \otimes (I_4 - P) + R \otimes P), \quad P = \frac{1}{4} \mathbf{11}^\top,$$

where the last tensor factors index the four blocks in (5.24), and

$$(5.26) \quad T = \begin{pmatrix} s & z & z & 2z \\ -y & 0 & 0 & -z \\ -y & 0 & 0 & -z \\ 2y & y & y & s \end{pmatrix}, \quad R = \begin{pmatrix} s & -3z & -3z & -6z \\ 3y & 2s & 2s & 3z \\ 3y & 2s & 2s & 3z \\ -6y & -3y & -3y & s \end{pmatrix}.$$

The square terms vanish for a structural reason. At the vacuum, $J_1(W) \subseteq \Lambda^0 E \oplus \Lambda^2 E$, whose Chevalley pairing with itself is zero in dimension eight. Hence $A_1 = 0$. The even particle-hole product sends 1 to vol , and the elementary creation product sends 1 to Ω_* . Lemma 5.4 gives $A_{\text{vol}} = A_{\Omega_*} = 0$. Thus the coefficients of $\alpha^2, \beta^2, \gamma^2$ in (5.19) are zero. The coefficients of $\alpha\beta, \alpha\gamma, \beta\gamma$ include both source orders. The only nonzero source subsets are unions of the four pairs B_r . On

(5.23), evaluating these three coefficients yields T ; on the central directions $(i, r), (j, s)$ it yields $T_{ij}\mathbf{1}_{r=s} + (R_{ij} - T_{ij})/4$; all other cross blocks vanish. These rules, together with the eight zero directions, specify all 120^2 entries. Pair-block permutations reduce the quartet substitution to the four parity choices for i, j ; their signs are precisely those in (5.23).

Since P and $I - P$ are complementary projections of ranks one and three, (5.25) gives, for every positive integer k ,

$$t_k(\eta) = 27 \operatorname{Tr}(T^k) + \operatorname{Tr}(R^k).$$

The trace computation can be performed with the following matrix identities, obtained by multiplying (5.26):

$$\begin{aligned} T^4 &= 2sT^3 - s^2T^2 + 4wT, \\ R^4 &= 6sR^3 - 9s^2R^2 + (4s^3 - 108w)R. \end{aligned}$$

The initial traces are

	$\operatorname{Tr}(\cdot)$	$\operatorname{Tr}((\cdot)^2)$	$\operatorname{Tr}((\cdot)^3)$
T	$2s$	$2s^2$	$2s^3 + 12w$
R	$6s$	$18s^2$	$66s^3 - 324w$

Multiplying each recurrence by I, T, T^2 or I, R, R^2 and taking traces gives (5.21)–(5.22). Equation (5.12) gives $\mathcal{B}(\eta, \eta) = 2(\alpha\beta + \alpha\gamma + \beta\gamma)$. Substituting these three expressions in (5.17) and collecting powers of s, w gives zero. All formulas are polynomial, so no parameter is required to be nonzero. $\square$

Lemma 5.6. *For*

$$v_A = e_0e_1e_2e_3, \quad v_B = e_4e_5e_6e_7, \quad \Psi = (1 + v_A)(1 + v_B),$$

one has

$$(5.27) \quad s_\Psi = 2, \quad t_4(\Psi) = 3648, \quad t_6(\Psi) = 57408, \quad P_{18}(\Psi) = 18063360 \neq 0.$$

Proof. Use (5.19), now with occupied-subset encodings 0, 15, 240, 255, all with coefficient one. The 64 directions between the two four-mode split spaces form an identity block. In each four-mode block, the twelve directions $u_i^+ \wedge u_j^-$ with $i \neq j$ give zero. Pair each $u_i^+ \wedge u_j^+$ with $u_k^- \wedge u_\ell^-$, where $k < \ell$ are the complementary two indices of that four-mode block. Its matrix is

$$\begin{pmatrix} 2 & 2(-1)^{i+j} \\ 2(-1)^{i+j} & 2 \end{pmatrix}.$$

There are twelve such two-dimensional blocks in total. Finally the four directions $u_i^+ \wedge u_i^-$ in each block have the all-ones matrix of size four. Entries between these blocks vanish by the same formula. Thus the eigenvalues of A_Ψ are 1 with multiplicity 64, 4 with multiplicity 14, and 0 with multiplicity 42. Hence $t_k(\Psi) = 64 + 14 \cdot 4^k$ for $k > 0$. The four complementary source pairs give $\mathcal{B}(\Psi, \Psi) = 4$. Substitution in (5.17) proves (5.27). $\square$

5.5. All three summands, including the boundary. The generic normal configuration in Lemma 5.5 has the common symplectic symmetry studied by Manivel [10, Proposition 1]. To exclude an arbitrary three-term decomposition, one must also treat third terms outside that nondegenerate chart. The following construction does so on the original vector by a polynomial specialization.

Lemma 5.7. *If g_1, g_2 are nonzero even pure spinors and $\mathcal{B}(g_1, g_2) \neq 0$, a common invertible transformation of the kinds in Lemma 5.4 sends them to $\alpha 1, \beta \operatorname{vol}$ with $\alpha\beta \neq 0$. The transformation acts simultaneously on any other summands.*

Proof. Choose a nonzero coefficient $(g_1)_I$. The set I is even. The product τ_I toggles precisely the occupation set I , so the vacuum coefficient of $\tau_I g_1$ is $\pm(g_1)_I \neq 0$. Apply its inverse chart unit from Lemma 5.2 to the whole configuration. The first term is now $\alpha 1$. The second term has top coefficient $\beta \neq 0$, since the common transformation has preserved $\mathcal{B}$.

Let $C = \tau_0 \cdots \tau_7$. The CAR give $C^2 = \text{Id}$ and $C1 = \text{vol}$, $C\text{vol} = 1$. The vector Cg'_2 has vacuum coefficient β , so $Cg'_2 = \beta \mathcal{U}1$ for a chart unit $\mathcal{U}$. Consequently $g'_2 = \beta C\mathcal{U}C \text{vol}$. Conjugation by C changes two-creation factors into two-contraction factors. Their inverse product fixes the vacuum and takes the second term to βvol . Apply that same product to all terms. $\square$

The first part of this proof also sends any even pure spinor to a nonzero vacuum multiple. Its self-pairing is therefore zero, by the nonzero form multiplier and $\mathcal{B}(1, 1) = 0$.

Proposition 5.8. *For every three nonzero even pure spinors $g_1, g_2, g_3 \in \mathcal{F}_8^+$,*

$$P_{18}(g_1 + g_2 + g_3) = 0.$$

Proof. Put $x = g_1 + g_2 + g_3$. If $s_x = 0$, the assertion follows from (5.17). Otherwise, since all self-pairings vanish, some pair has nonzero pairing. By Lemma 5.7 and relative covariance it suffices to consider

$$x = \alpha 1 + \beta \text{vol} + s,$$

where s is an arbitrary nonzero even pure spinor.

Choose a nonzero coefficient s_I , order I increasingly, and divide it into $k = |I|/2$ successive pairs (i_r, j_r) with $i_r < j_r$. Define the actual polynomial curve

$$(5.28) \quad h(t) = \prod_{r=1}^k (\text{Id} + t \iota_{f_{j_r}} \iota_{f_{i_r}}) s, \quad a(t) = [h(t)]_0.$$

The factors commute, square-zero generators give their inverses by $t \mapsto -t$, and their Clifford shears preserve purity. Thus every $h(t)$ is nonzero and pure. The coefficient of t^k in $a(t)$ is exactly s_I : all chosen contractions produce a vacuum only from the complete original support I , and the successive ordered-pair convention gives sign $+1$. Hence $a(t)$ is a nonzero polynomial. If $k = 0$, this statement simply reads $a(t) = s_\emptyset \neq 0$.

Put $\omega_* = e_0 e_1 + e_2 e_3 + e_4 e_5 + e_6 e_7$ and

$$(5.29) \quad r(t, u) = \prod_{j=0}^3 (\text{Id} + u \varepsilon_{2j} \varepsilon_{2j+1}) h(t).$$

This is again nonzero and pure for every (t, u) , and $r(0, 0) = s$. Its first two even components are

$$[r(t, u)]_0 = a(t), \quad [r(t, u)]_2 = [h(t)]_2 + u a(t) \omega_*.$$

Let $K(\omega) : E^* \rightarrow E$ be $f \mapsto \iota_f \omega$ and $J = K(\omega_*)$, an invertible alternating matrix. The polynomial

$$(5.30) \quad \Delta(t, u) = a(t) \det(K([h(t)]_2) + u a(t) J)$$

is nonzero: its coefficient of u^8 is $a(t)^9 \det J$, nonzero in $\mathbb{C}[t]$. At every point where $\Delta \neq 0$, the vacuum coefficient is nonzero and (5.7) gives a nondegenerate graph

$$\mathcal{D} = -K([r(t, u)]_2 / a(t)).$$

The symplectic-basis construction after (5.10) gives a $g \in \text{GL}(E)$ taking it to the graph of Ω_* . Lemma 5.2 and graph uniqueness then give $\Lambda g r(t, u) = a(t) \Omega_*$. The same Λg fixes 1 and multiplies vol by $\det g$. It therefore sends the entire sum $\alpha 1 + \beta \text{vol} + r(t, u)$ into the family of Lemma 5.5. Relative covariance proves

$$F_{16}(\alpha 1 + \beta \text{vol} + r(t, u)) = 0 \quad \text{whenever } \Delta(t, u) \neq 0.$$

The left side is a polynomial $P(t, u)$. The product $P\Delta$ vanishes at every point of $\mathbb{C}^2$, hence is the zero polynomial: a polynomial over an infinite field is determined by all of its evaluations, by applying the one-variable root bound successively to the two variables. Since $\mathbb{C}[t, u]$ is a domain and $\Delta \neq 0$, we have $P = 0$. Evaluating at $(0, 0)$ recovers the exact original s , so $F_{16}(\alpha 1 + \beta \text{vol} + s) = 0$. There is no limiting assertion about the existence of a decomposition; only the vanishing of one already defined polynomial is extended. $\square$

Corollary 5.9. *The vector $\Psi = (1 + v_A)(1 + v_B)$ has Gaussian rank four.*

Proof. Its four original occupation terms give the upper bound. A representation with at most three Gaussian terms is a representation by even pure spinors. If necessary, split one nonzero term into two nonzero scalar multiples to obtain exactly three terms. Proposition 5.8 would then give $P_{18}(\Psi) = 0$, contrary to Lemma 5.6. $\square$

Example 5.10. Each vector

$$\prod_{i=0}^3 (1 + z_i e_i e_{i+4}), \quad z_i \in \mathbb{C},$$

is an eight-mode Gaussian with pairing across the two original four-mode blocks. Such vectors are legitimate summands in a candidate shorter expansion of Ψ . The preceding argument does not restrict them to block products. It transports the whole proposed sum by one common map; independently normalizing its three summands would destroy the sum identity and would not imply the polynomial obstruction.

5.6. Arbitrary four-mode factors.

Lemma 5.11. *Every nonzero four-mode even vector is either Gaussian, of rank one, or is carried to a nonzero scalar multiple of $1 + \text{vol}_4$ by an invertible transformation preserving the Gaussian cone. In the latter case its rank is two. The required transformations can be chosen as complex Spin transformations followed by nonzero scalars.*

Proof. An even particle-hole product makes a nonzero occupation coefficient a nonzero vacuum coefficient a . Write the resulting vector as

$$v = a1 + \omega + b\text{vol}_4, \quad \omega \in \Lambda^2 \mathbb{C}^4.$$

Choose the six elementary creation factors whose degree-two part is ω/a , and call their product $\mathcal{U}$. There is a scalar d such that $\mathcal{U}1 = 1 + \omega/a + d\text{vol}_4$. Positive-degree creators kill the volume, so

$$\mathcal{U}^{-1}v = a1 + c\text{vol}_4, \quad c = b - ad.$$

If $c = 0$, the vector is Gaussian by (5.5). If $c \neq 0$, it is not pure by Example 5.3, and has rank two. A diagonal $g \in \text{GL}_4(\mathbb{C})$ with $\det g = a/c$ takes it to $a(1 + \text{vol}_4)$.

Here are explicit Spin lifts for all the operations, with split directions u_i^+, u_i^- defined as above in four modes. A particle-hole operator is $c(u_i^+ + u_i^-)$, and $q(u_i^+ + u_i^-) = 1$, so an even product belongs to Spin. For an elementary creation unit choose k distinct from i, j and put $v = u_k^+ + u_k^-$ and $w = v + u_j^+$. Both have quadratic norm one and are orthogonal to u_i^+ . In the Clifford algebra,

$$(v - zu_i^+)v(w + zu_i^+)w = (1 - zu_i^+v)(1 + zu_i^+w) = 1 + zu_i^+u_j^+.$$

All four vector factors have quadratic norm one. This explicitly realizes $1 + z\varepsilon_i\varepsilon_j$ as a Spin action. For the last diagonal scaling, choose $r \in \mathbb{C}^*$ with $r^2 = a/c$. On one mode let $P_0 = \iota_i\varepsilon_i$ and $P_1 = \varepsilon_i\iota_i$. The two Clifford vectors $ru_i^+ + r^{-1}u_i^-$ and $u_i^+ + u_i^-$ have quadratic norm one. Their product acts as $r^{-1}P_0 + rP_1$ and conjugates creation and contraction in that mode by factors r^2, r^{-2} . Its spin action is $r^{-1}\Lambda g$ for the diagonal g used above. This supplies the claimed Spin transformation up to a nonzero scalar without using a classification of half-spin orbits. $\square$

Proof of Theorem 5.1. First suppose both four-mode factors are non-Gaussian. Lemma 5.11 gives local complex Spin operations and scalars taking them to $1 + \text{vol}_4$. For the orthogonal direct sum $U_A \oplus U_B$, the natural map

$$(5.31) \quad \text{Spin}(U_A) \times \text{Spin}(U_B) \longrightarrow \text{Spin}(U_A \oplus U_B), \quad (x, y) \longmapsto xy,$$

is a homomorphism because the even Clifford subalgebras commute. It is not asserted to be injective: $(-1, -1)$ maps to the identity. On the even-even Fock tensor factor its action is the usual tensor action. Thus the local operations induce one invertible eight-mode transformation preserving the entire physical Gaussian cone by (5.5). They take $\psi \otimes \phi$ to a nonzero multiple of Ψ . Corollary 5.9 gives rank four, which is the product of the two local ranks.

If one factor is Gaussian, use a local compact Gaussian operation and a scalar to make it the vacuum. Tensoring a minimal decomposition of the other factor with that vacuum gives one inequality. For the reverse inequality, take any eight-mode Gaussian decomposition of $1_A \otimes \phi$ and apply the same vacuum coefficient functional on all four modes of A to every summand. This is the composition of their vacancy projections followed by removal of those vacuum modes. The nonzero projected terms remain pure by the isotropic Clifford argument following (5.5). To see that removal preserves purity, write a nonzero projected vector as $1_A \otimes t_B$. The A -degree-one and A -degree-zero parts of its Clifford equation are independent, so

$$\text{Ann}(1_A \otimes t_B) = E_A^* \oplus \text{Ann}(t_B).$$

Its dimension is eight, hence $\dim \text{Ann}(t_B) = 4$. Parity is even, and (5.5) makes t_B a physical Gaussian vector. After discarding zero terms, the projected sum is a Gaussian decomposition of the exact original ϕ , with no more terms. This proves equality of ranks. When both factors are Gaussian, their tensor product is Gaussian by the compact version of (5.31). All cases of (5.2) follow. $\square$

Corollary 5.12. *For the four-mode state*

$$\widetilde{M} = \frac{\sqrt{3}|0000\rangle + \sqrt{2}|0011\rangle + \sqrt{2}|1100\rangle + |1111\rangle}{\sqrt{8}},$$

one has $\chi_G(\widetilde{M}) = 2$ and $\chi_G(\widetilde{M}^{\otimes 2}) = 4$.

Proof. Before normalization its degree-two part is the sum of two disjoint pair terms, each with coefficient $\sqrt{2}$. Their product has top coefficient 2, whereas the product of its vacuum and top coefficients is $\sqrt{3}$. Equivalently, the reduction in Lemma 5.11 gives $c = 1 - 2/\sqrt{3} \neq 0$. The local rank is therefore two, and Theorem 5.1 gives the product rank. $\square$

This state also occurs in the approximate-decomposition experiments of [5]. The corollary concerns exact rank; it does not determine the minimum number of Gaussian terms at a positive approximation error.

The current Lean development verifies the original rank bounds $2 \leq \chi_G(\Psi) \leq 4$ and the complete finite-coordinate identities above; it does not yet formalize the rank-four theorem end to end. The precise verified statements and remaining identifications are described in Section 7.

6. ENTROPY DEFICITS AND COHERENT MEASUREMENT CAPACITY

The distribution comparison also gives quantitative information about the input state and the measurement it defines. The first application relates the Wehrl entropy deficit to fractional occupation numbers. The second determines the mutual-information capacity of the coherent measurement, including repeated measurements on entangled inputs.

6.1. An occupation-number bound for the entropy deficit.

Proposition 6.1. *Let $2 \leq p \leq n$, let $\psi \in \Lambda^p \mathbb{C}^n$ be a unit vector, write $\rho = |\psi\rangle\langle\psi|$, and put $D = \binom{n}{p}$. With the one-particle matrix γ_ρ from (2.8), and the coherent moments and entropy constants of Section 4,*

$$(6.1) \quad S_W(\rho) - s_{p,n} \geq \frac{D m_{p-1,n-1}(2)}{2n(n+1)} \operatorname{Tr}[\gamma_\rho(\operatorname{Id} - \gamma_\rho)].$$

Proof. The original occupation conditioning (2.13), followed by the lower-sector comparison with the test t^2 , gives

$$(6.2) \quad \int Q_\rho^2 d\nu_{p,n} \leq m_{p-1,n-1}(2) \mathbb{E}_u \langle u, \gamma_\rho u \rangle^2,$$

where u is uniform on the complex unit sphere. Zero conditional branches contribute zero, so this formula also covers singular ρ and vanishing occupation probabilities.

For every Hermitian matrix A , diagonalization and unitary invariance give

$$(6.3) \quad \mathbb{E}_u \langle u, Au \rangle^2 = \frac{(\operatorname{Tr} A)^2 + \operatorname{Tr}(A^2)}{n(n+1)}.$$

Indeed, normalize a vector of independent standard complex Gaussian coordinates. The squared moduli of its normalized coordinates are uniform on the probability simplex. Integrating its monomials gives $\mathbb{E}|u_i|^4 = 2/[n(n+1)]$ and $\mathbb{E}|u_i|^2|u_j|^2 = 1/[n(n+1)]$ for $i \neq j$. Expanding the square in an eigenbasis proves (6.3); the formula also holds for $n = 1$.

A coherent Slater input has one-particle matrix π_W , with $\operatorname{Tr} \pi_W = \operatorname{Tr}(\pi_W^2) = p$. Its conditional states are Slater, so (6.2) is an equality for that input. Since $\operatorname{Tr} \gamma_\rho = p$, subtraction yields

$$(6.4) \quad m_{p,n}(2) - \int Q_\rho^2 d\nu_{p,n} \geq \frac{m_{p-1,n-1}(2)}{n(n+1)} \operatorname{Tr}[\gamma_\rho(\operatorname{Id} - \gamma_\rho)].$$

Finally, $t \log t - \frac{1}{2}t^2$, extended continuously at zero, is convex on $[0, 1]$, since its second derivative on $(0, 1]$ is $t^{-1} - 1 \geq 0$. Theorem 2.1 applied to this function gives

$$S_W(\rho) - s_{p,n} \geq \frac{D}{2} \left(m_{p,n}(2) - \int Q_\rho^2 d\nu_{p,n} \right).$$

Combining this with (6.4) proves the result. $\square$

The right-hand side sums $\lambda_j(1 - \lambda_j)$ over the natural occupation numbers. It vanishes exactly when the one-particle matrix is a projection, which by the equality argument of Theorem 2.1 forces a pure Slater state. The coefficient in (6.1) is not asserted to be optimal. For $p = n$ the sector is one-dimensional and both sides are zero.

6.2. The information capacity of the coherent POVM. Fix one of the coherent orbits considered above. Let $\mathcal{H}$ be its representation space, $D = \dim \mathcal{H}$, ν its invariant probability, and s_* its coherent Wehrl entropy from Theorem 4.2. The resolution of the identity defines the positive operator-valued measure

$$(6.5) \quad E(dx) = D|\Omega_x\rangle\langle\Omega_x| d\nu(x).$$

An input density operator ρ produces the density $f_\rho(x) = DQ_\rho(x)$ relative to ν . The *measurement capacity* is the supremum of $I(Z : X)$ over measurable input ensembles $z \mapsto \rho_z$ with a probability law for the classical label Z ; the measurement itself remains fixed.

Proposition 6.2 (Capacity and repeated inputs). *For the POVM (6.5),*

$$(6.6) \quad C_{\text{meas}} = \log D - s_*.$$

For every positive integer ℓ , its ℓ -fold product measurement has capacity $\ell(\log D - s_)$, allowing arbitrary joint input ensembles of density operators on $\mathcal{H}^{\otimes \ell}$. In particular, this assertion includes entangled inputs.*

Proof. For a probability density f with respect to the probability measure ν , write $h_\nu(f) = -\int f \log f d\nu$. The normalization of Wehrl entropy gives

$$(6.7) \quad h_\nu(f_\rho) = S_W(\rho) - \log D \geq s_* - \log D.$$

Also $h_\nu(f) \leq 0$, by convexity of $t \log t$ and $\int f d\nu = 1$. For an ensemble $(\rho_z, \pi(dz))$, put $\bar{f} = \int f_{\rho_z} \pi(dz)$. Then

$$I(Z : X) = h_\nu(\bar{f}) - \int h_\nu(f_{\rho_z}) \pi(dz) \leq \log D - s_*.$$

These entropy expressions are finite because all measurement output densities are bounded by D . Equality is attained by the invariant ensemble of pure coherent projectors: its average input is Id/D , its average output density is one, and each input has Wehrl entropy s_* . A finite ensemble can attain the same value. Indeed, Id/D belongs to the convex hull of the compact orbit of coherent projectors, and finite-dimensional convexity gives a finite convex combination with this average; every member still has the same minimum output entropy.

For repeated measurements, fix any density operator ρ on $\mathcal{H}^{\otimes \ell}$. Its joint output density is

$$f_\rho(x_1, \dots, x_\ell) = D^\ell \text{Tr} \left[\rho \bigotimes_{j=1}^{\ell} |\Omega_{x_j}\rangle \langle \Omega_{x_j}| \right].$$

For a history $(x_1, \dots, x_{j-1})$ of positive marginal density, define

$$K = \bigotimes_{i < j} \sqrt{D} |\Omega_{x_i}\rangle \langle \Omega_{x_i}|, \quad \tau = \frac{\text{Tr}_{1, \dots, j-1}[(K \otimes \text{Id})\rho(K^* \otimes \text{Id})]}{\text{Tr}[(K \otimes \text{Id})\rho(K^* \otimes \text{Id})]}.$$

The operator τ is a density operator on the remaining factors. Its first reduced density operator, say τ_j , produces the conditional density of x_j by the same POVM: $f(x_j | x_1, \dots, x_{j-1}) = D Q_{\tau_j}(x_j)$. This follows by inserting the resolution of the identity in the unmeasured factors and taking the indicated partial trace. Histories with zero marginal density make no contribution.

Each conditional density is bounded by D and satisfies (6.7). The joint density is bounded by D^ℓ . Thus $f \log f$ is integrable for each joint and conditional law; in particular, its negative part is bounded in integral by $1/e$ and its positive part by the corresponding logarithmic bound. The entropy chain rule is therefore valid and gives

$$(6.8) \quad h_{\nu^{\otimes \ell}}(f_\rho) = \sum_{j=1}^{\ell} \mathbb{E} h_\nu(f(\cdot | X_1, \dots, X_{j-1})) \geq \ell(s_* - \log D).$$

Product coherent inputs attain this bound. For an arbitrary joint input ensemble, the entropy of its average output relative to $\nu^{\otimes \ell}$ is at most zero. The mutual-information identity therefore gives the upper bound $\ell(\log D - s_*)$. Independent copies of an optimal coherent ensemble attain it, which proves the capacity formula even with joint encoding. $\square$

7. FORMAL VERIFICATION

The Lean development accompanies the proofs above and records which parts have been checked on their original mathematical objects. Its source and verification evidence are available in the repository [11]. The verified source snapshot is [3164a3](#); the English manuscript was added subsequently without changing those proof sources.

7.1. End-to-end exterior-power theorems. The formalization of Theorem 2.1 starts with $\Lambda^p \mathbb{C}^n$, its actual unitary action, density operators, and normalized Haar measure. It constructs creation and contraction, the conditional density on the orthogonal complement, the double-Haar identity, and the common occupation mixture. The induction and the strict equality argument are proved from these constructions. In particular, the statement covers every $0 \leq p \leq n$, all mixed inputs, and every continuous convex test on $[0, 1]$; a strictly convex test has equality exactly at pure Slater densities. No conditional-measure identity or full-density rigidity statement is assumed as a separate hypothesis.

The formal Slater measure is the pushforward to the actual orbit of rank-one Slater projectors. Its invariance, independence of the reference, and agreement with group integration are proved. The entropy results then use this same measurement. They include the sector dimension $\binom{n}{p}$, the normalized first moment, all positive real moments, the Wehrl minimum, and the Rényi–Wehrl minimum for every real $q > 0$, $q \neq 1$, together with the same equality characterization. These minimizer theorems are end to end. Their coherent comparison values are defined by integrals; the Beta-product evaluations, explicit Gamma products, harmonic-number constants, and the limit as $q \rightarrow 1$ have not been formalized.

The principal declarations are listed below. Each namespace has the prefix `Fermionic.`; the repository prints the full types of all declarations, including their quantifiers and domain assumptions.

Namespace	Main declarations
<code>HusimiInduction</code>	<code>exterior_husimi_convex_order</code> <code>exterior_husimi_equality_iff</code>
<code>SlaterMeasure</code>	<code>orbit_husimi_convex_order</code> <code>orbit_husimi_equality_iff</code>
<code>ExteriorEntropy</code>	<code>wehrl_minimum</code> , <code>wehrl_eq_iff_slater</code> <code>renyiWehrl_minimum</code> <code>renyiWehrl_eq_iff_slater</code>

7.2. The Gaussian-rank boundary. For the actual even Spin-vacuum-orbit Gaussian dictionary on eight modes, the formalized results give, for the original occupation vector Ψ defined in Section 5,

$$2 \leq \chi_G(\Psi) \leq 4.$$

They include the actual Clifford representation, the four-term Gaussian decomposition, purity of Gaussian orbit vectors, the vacuum-chart annihilator calculation, and nonpurity of the target. Formal scale invariance gives the same interval for $\Psi/2$. Its occupation-space identification with $M^{\otimes 2}$ is made in the paper; a separate Hilbert tensor-product identification is not part of the verified roots. Separately, the full normal-family certificate in Section 5 is checked for arbitrary complex parameters, including zero parameters. The encoded target has the nonzero certificate value 18063360.

These results do not yet constitute an end-to-end formalization of $\chi_G(M^{\otimes 2}) = 4$. The remaining Lean work is the original Fock-to-coordinate identification, covariance of the certificate under the actual common transformations, and the reduction of every three-term pure-spinor sum, including its degenerate cases. Section 5 gives the mathematical argument for these steps. The half-spin and odd-spin convex-order theorems, their entropy consequences, and the applications in Section 6 likewise have paper proofs here, rather than completed end-to-end Lean proofs. The repository's `TODO.md` records the outstanding formal statements and their paper-level derivations.

7.3. Verification procedure and foundations. The recorded runs use Lean 4.29.0 and mathlib at revision 8a178386ffc0f5fef0b77738bb5449d50efeea95. All Lean and Lake execution took place on a CAB server through SSH. Three suites passed clean project compilation with warnings

treated as errors, inspection of all selected declaration types and their transitive axioms, and a fresh kernel replay of the imported proof environment. The exterior suite contains 20 modules and 309 theorems; the rank foundation and certificate suite contains 33 modules and 257 theorems; the vacuum-chart suite contains 12 modules and 150 theorems. These suites overlap and were verified in three separate runs. Their shared source files are identical.

The proofs use no `sorry`, `admit`, added mathematical axiom, or `native_decide`. Finite certificate leaves use ordinary kernel-checked decision procedures, and parameter identities use kernel-checked algebraic proofs. The disclosed logical axioms are `propext`, `Classical.choice`, and `Quot.sound`. Fresh replay starts from an empty environment and rechecks the replayable safe, non-partial constants with the same Lean kernel; axioms remain axioms. It is not an independent implementation of the kernel. The dependency versions are pinned, but their source files were not all recompiled from scratch in each run. Source hashes, complete logs, and reproduction instructions are included in the repository.

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